# USER MANUAL

Machine: EL8A Fiber Laser Cutting Machine

Manufacturer: Foshan Huibaisheng Laser Technology Co.,Ltd.

Add.: No.10 Hexin Road, Yanghe Town, Gaoming District, Foshan City, Guangdong Province, China

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# Foreword

Thank you for choosing the Baison Laser Cutting Machine. As a leader in laser cutting technology, Baison Laser is committed to providing high-efficiency, precision, and reliable metal processing solutions to customers worldwide. With rigorous quality control and innovative design, our machines not only enhance production efficiency but also deliver significant cost savings and competitive advantages for your business.

This manual is designed to help you quickly understand the operation and maintenance of the equipment, ensuring optimal performance. Please read carefully and follow the operational guidelines provided to maximize the machine's potential. The equipment manual is intended not only for the machine's users (owners) and operators but also for maintenance personnel. All of these individuals should have access to it. Should you have any questions, feel free to contact our technical support team, who will be happy to assist you.

## Version Information

This version provides detailed information on the technical specifications, installation and commissioning, operating instructions, maintenance and upkeep, software usage, and troubleshooting solutions for the equipment. In the event of any product improvements or feature updates in the future, we will promptly release a new version of the manual to ensure you have access to the latest product information and operating guidelines.

**Version History**

| Version Number | Release Date |
|---|---|
| V1.5.1 | 2025-11 |
| | |

## Manufacturer Information

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# Disclaimer

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## Notice

**1. Operational Risks**

The fiber laser cutting machine is only to be operated by personnel who have received professional training. Operators must thoroughly read and understand all contents of this manual and strictly follow the operating instructions and safety precautions.Our company is not responsible for any damages or malfunctions caused by improper operation, unauthorized modifications or repairs, or failure to maintain and upkeep the equipment as instructed in this manual.

**2. Limitation of Liability**

Our company assumes no liability for any indirect, special, incidental, or consequential damages arising from the use of this equipment, including but not limited to production losses, loss of profits, or data loss.

**3. Scope of Application**

The information in this manual applies only to the specified model of the fiber laser cutting machine. Any unauthorized modifications to the equipment will void the warranty.

**4.Data and Information**

The technical data and information in this manual may be updated as the product is improved, and the manufacturer reserves the right to make modifications without prior notice.

We hope this manual provides you with valuable guidance, ensuring you can use the fiber laser cutting machine safely and effectively.If you have any questions or require further technical support during use, please feel free to contact our company's customer service center.Thank you for choosing our company's products, and we will continue to provide you with high-quality service.

Foshan Huibaisheng Laser Technology Co., Ltd.
Jan. 1, 2024

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# Chapter 1 Safety

This chapter describes the safety concept. The chapter describes how potential dangers can be avoided. The overview of residual risks contains measures to be taken by the operator in order to reduce the residual risks.

Note：
Operators must comply with the valid safety and accident prevention regulations of their respective countries, as well as national and regional safety laws!

## 1.1 Safety Regulations and Guidelines

GB7247.1-2012 Safety of Laser Products, Part 1: Equipment Classification and Requirements
Compliance with GB10320-2011 for Electrical Safety of Laser Equipment and Facilities

## 1.2 Safety Precautions

To ensure the safe operation of the fiber laser cutting machine and prevent any accidents or injuries, strictly adhere to the following safety precautions and maintain a high level of safety awareness. Regular safety training should be conducted to ensure that all operators are familiar with safety protocols and can respond appropriately to potential hazards. Through strict safety management and standardized operation, accidents can be effectively prevented, ensuring personnel safety and the normal operation of the equipment.

### 1.2.1 Laser Radiation Hazards

**Direct radiation:** The laser beam emitted by the fiber laser has an extremely high energy density, which can cause serious damage to the eyes and skin. During operation, protective laser goggles that meet the required standards must be worn, and the laser beam should be operated within a closed or isolated space.
**Reflected radiation:** Reflective surfaces can cause laser beam scattering, which is equally hazardous. The operating environment should avoid high-reflective materials, and appropriate protective equipment should be worn.

**Protective measures:** Set up laser safety barriers to ensure that only authorized personnel enter the laser work area, and post prominent laser warning signs in the operational area.

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### 1.2.2 Electrical Hazards

**High-voltage power:** The laser cutting machine uses a high-voltage power supply, and improper operation may lead to electric shock accidents. Installation and maintenance work should be performed by qualified professionals.

**Wires and connections:** Regularly check wires and connections for damage or looseness to prevent electrical short circuits. Do not operate the equipment in a damp environment.

**Emergency power shutdown measures:** Ensure operators are familiar with emergency power shutdown procedures and the location of the emergency power-off button to quickly cut off the power in an emergency.

### 1.2.3 Mechanical Hazards

Moving parts: The worktable and robotic arm of the laser cutting machine move at high speeds during operation, so do not touch any moving parts while the equipment is running.

Clamping devices: Ensure the workpieces are securely clamped to prevent them from coming loose and causing injury during processing.

Maintenance and servicing: Before performing maintenance and servicing, the machine must be fully powered off, and all moving parts must come to a complete stop before any operation is conducted.

### 1.2.4 Fire and Explosion Hazards

Flammable materials: The high temperatures generated during laser cutting may ignite flammable materials. The work area should be kept away from combustible materials and equipped with proper fire extinguishing devices.

Ventilation and smoke extraction: Ensure that the work area is well-ventilated and that the exhaust system functions properly to remove smoke and dust produced during cutting, preventing fire hazards.

Operational procedures: Strictly adhere to operational procedures and avoid overloading the equipment to prevent fire risks caused by overheating.

### 1.2.5 Noise and Vibration

Noise control: The laser cutting machine produces high-decibel noise during operation. Operators should wear protective earplugs or earmuffs that meet safety standards to prevent hearing damage.

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Vibration impact: The vibrations generated during operation can affect cutting precision. Ensure that the equipment is installed on a stable foundation and regularly check for proper tightening.

### 1.2.6 Chemical Hazards

Harmful gases: Cutting certain materials may release harmful gases, such as plastics or coated metals. Ensure the work area is well-ventilated or use professional smoke extraction devices.

Cleaning agents and lubricants: When using chemical cleaning agents and lubricants, follow the manufacturer's safety instructions, wear appropriate protective gloves and goggles, and avoid skin contact and inhaling harmful fumes.

Waste disposal: Collect and dispose of waste and residues from cutting processes in accordance with environmental regulations to avoid pollution and chemical hazards.

## 1.3 Safety Warning Signs and Instructions

Before using the laser cutting machine, please be sure to carefully read all safety information in this manual and the safety warning signs on the machine. These signs are intended to alert operators to potential hazards and guide safe operation. Ensure that these safety signs are always kept in place, in good condition, and clearly visible.

If any safety signs are found to be damaged, faded, or missing, they must be replaced immediately. Please note that new safety signs should be strictly affixed in their original positions, and it is forbidden to arbitrarily change or remove the signs. Additionally, some signs may include signal words such as "Danger" or "Warning" to specifically alert to high-risk areas and operations that may cause serious injury.

Safety label stickers are typically located in specific hazardous areas on the machine, such as the laser generator, moving parts, and cutting areas.
Operators must follow these safety warnings, avoid direct contact with hazardous areas, and wear appropriate protective gear during all operations to ensure personal safety.

| Signal Words | Explanation |
|---|---|
| Danger | ...Indicates the most serious hazardous situation, with direct risk of severe injury or death. |
| Warning | ...Indicates a potential hazardous situation that, if not avoided, could result in very serious injury or significant property damage. |
| Caution | ...ndicates a potential hazardous situation that, if not avoided, could result in minor injury or property damage. |

Table 1-1

| | |
|---|---|
|  | Protective goggles must be worn |
|  | Danger due to laser radiation |
|  | Read instructions before use |
|  | Beware of pinching |
|  | This machine may generate toxic fumes/particles |
|  | Caution: Class 2M laser radiation when opened. Do not look directly at the beam or view directly with optical instruments. |
|  | Danger: Class 4 laser radiation when opened. Avoid direct or scattered exposure to eyes or skin. |
|  | Risk of electric shock |
|  | Beware of mechanical injury |
|  | Do not approach while the machine is running |

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| | |
|---|---|
|  | This equipment must only be operated and maintained by qualified personnel |
|  | Always wear protective goggles when operating the machine |

Table 1-2

## 1.4 Operational Safety

If the machine is used improperly or for unintended purposes, it may become a source of danger:
- Endangering the safety of operators.
- Damaging the machine and the operator's other physical assets.
- Negatively impacting the normal operation of the machine.

### 1.4.1 The Machine

This equipment is restricted to industrial use. The installation, operation, and transportation conditions specified by Baison Laser must be followed, and regular maintenance performed on time. The installation and operation of the equipment must comply with the relevant regulations of the country in which it is located. Users must strictly adhere to national regulations.

Operators may use the equipment for laser cutting of metal sheets. If the equipment is equipped with a pipe cutting unit, it can also be used for laser cutting of metal pipes and profiles.

The following behaviors are prohibited:
- Unauthorized changes or modifications to the machine by users or personnel.
- Any work procedures that affect safety.
- Laser cutting of
  – Plastic.
  – Wood.
  – Sheets coated with PVC foil.
  – Magnesium.

**Disclaimer**
Any other form of use is considered unauthorized. Baison Laser assumes no responsibility for any damages resulting from this (especially injuries and production failures). The operator bears all risks, and the machine warranty will be voided.

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## 1.4.2 Authorized Personnel

The operation, setup, and maintenance of the equipment are restricted to authorized, trained, and instructed personnel. Qualified personnel may perform the following tasks:

- ✓ Transport the equipment to the installation site.
- ✓ Operate the laser-related modules, hydraulic modules, pneumatic modules, and electrical modules.
- ✓ Disassemble the equipment.

## 1.4.3 Precautions

- ➢ Ensure good indoor lighting in the operating area so the operator can work smoothly.
- ➢ Before running the equipment, check if there are any personnel lingering in the surrounding area to prevent accidents when starting the machine.
- ➢ Equipment operators or those near the laser during its use must wear laser protective glasses suitable for 1064nm wavelength protection.
- ➢ It is strictly forbidden to look directly at the laser through non-protective windows without wearing protective glasses.
- ➢ To protect operators, the equipment must be equipped with processing chambers or protective barriers to prevent laser leakage.
- ➢ Opening the processing chamber door should close the laser shutter.
- ➢ When entering the working area of the equipment, the servo button must be turned off to prevent accidental startup of the equipment and potential safety accidents.
- ➢ When not using the laser cutting machine, the machine's key (and the laser key if separately equipped) should be removed to prevent unauthorized personnel from operating the machine.
- ➢ The laser head, cut workpieces, and cutting platform are high-temperature components that must be cooled before touching. Direct hand contact with cut parts is strictly prohibited.
- ➢ Materials that are not heat-resistant or unsuitable for laser processing must not be laser cut to avoid endangering the operator and the equipment.
- ➢ New workpieces must undergo test runs, and their working conditions should be checked.
- ➢ When the equipment starts operating, the operator must closely monitor the cutting status and must not leave the station without authorization. If necessary to leave, the machine must be stopped, or the power switch turned off.
- ➢ During operation, it is strictly forbidden to change equipment parameters arbitrarily to avoid causing malfunctions.

| | |
|---|---|
|  | It is recommended to use gloves made of heat-resistant materials.The safety level of the gloves should comply with international standards, such as EN407 heat-resistant gloves, to ensure adequate protection. |

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| | |
|---|---|
|  | The wavelength of fiber lasers is typically 1064 nm; ensure the glasses can protect against this wavelength. |
|  | If the operating environment has a high concentration of harmful particles, FFP2 or FFP3 masks are recommended; FFP3 filters at least 99% of particles for better protection. |

## 1.5 Hazards

### 1.5.1 Laser Classification

Laser equipment is classified based on the potential hazard the laser poses to the human eye and skin. The international standards for laser classification are based on IEC 60825-1 (International Electrotechnical Commission) and GB7247.1. Below is a detailed description of the laser classification system:

| Laser Class | Description |
|---|---|
| Class 1 | Safe for human exposure. Even with prolonged exposure, it does not pose any harm to the eyes or skin. |
| Class 1M | Generally safe, but may cause eye injury if viewed directly through optical devices such as magnifiers or telescopes. |
| Class 2 | Emits visible light with an output power of less than 1 mW (in the range of 400nm to 700nm). Brief exposure will not harm the eyes due to the body's natural aversion response (e.g., blinking). |
| Class 2M | Similar to Class 2, but may be hazardous if viewed through optical devices such as telescopes. |
| Class 3R | Low-power laser, outputting less than 5 mW. Although not likely to cause injury from brief exposure, direct viewing is discouraged. |
| Class 3B | Emits laser output power between 5 mW and 500 mW. Direct exposure to the eyes can cause severe injury. Skin damage is possible, but less significant. Protective measures are necessary. |
| Class 4 | Outputs more than 500 mW. It poses serious risks to the eyes and skin, and may cause fires. Strict protective measures are required. |

Table 1-3

### 1.5.2 Laser Classification of This Machine

The machine contains a fiber laser of laser class 4. — the highest and most hazardous laser class. Class 4 lasers

have extremely high output power, capable of causing serious injuries to both eyes and skin, and they may even ignite fires.

Because the laser cutting machine uses a high-power laser beam to cut through metals and other materials, significant safety risks are present during operation. Therefore, laser cutting machines must be equipped with protective enclosures, laser shielding, and other safety measures. Operators must wear appropriate laser safety goggles and strictly adhere to safety protocols.

Class 4 laser systems are generally equipped with emergency stop buttons and interlock systems to ensure that if the machine housing is opened, the laser will automatically cease operation, thereby preventing accidental injury.

The machine/safety light barrier can contain helium-neon laser or laser diodes with red light for adjustment. They correspond to laser class 2.

## 1.5.3 Overview of residual risks

The machine has residual risks in spite of its safety devices and construction type .

| Residual risks | Dangerous point | Type of danger | Measure to be taken by the operator |
|---|---|---|---|
| **Mechanical elements** | | | |
| Squeezing or impact | Movement of the X, Y and Z axes and the pallet, conveyor belts | Risk of injury | |
| | Loading and unloading cart (option) | Risk of injury | |
| | Storage system (option) | Risk of injury | |
| Shearing | Raising, lowering, and traveling of the pallet | Risk of injury | |
| Cutting and cutting to size | Sharp-edged workpieces. | Risk of injury | Wearing protective clothing and gloves.<br>Use of tools to remove workpieces. |
| By interception or coiling | Movements of workpieces projecting out of the machine | Risk of injury | |
| | Chain drive of the pallet changer | Risk of injury | |
| Due to being drawn in or caught | Pallet movement on uppertrack. | Risk of injury | |

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| | | | |
|---|---|---|---|
| Due to fluids and gases being expelled under high pressure | Cutting gas | Risk of injury | |
| **Electrics** | | | |
| Electrical contact | Directly with parts that are normally live. | Risk of fatal injury. | |
| | Indirectly with parts that are live due to defects. | Risk of fatal injury. | |
| | Contact with parts storing electrical charges | Risk of fatal injury. | |
| **Heat** | | | |
| Due to thermal radiation or spurting, melted parts. | Ejected slag spatters | Risk of injury | |
| Due to touching | Hot workpieces | Risk of injury | |
| Due to flames or explosion | Fires/detonations due to concentration of oxygen | Risk of injury | |
| **Secondary radiation (extremely bright and intensive light at the machining point)** | | | |
| Eye contact | Due to plasma formation, intense, visible radiation at the machining point. | Risk of injury Permanent damage to the retina. | • Personal eye protection: wear glasses with darkened lenses or hold a darkened plate in front of your eyes.<br>• Observe national regulations. |
| **Materials** | | | |
| Through contact with or by inhaling toxic fluids, gases, mist, vapors, and dust. | Cutting gas, dust, aerosols, cutting oils, PE-coated profiles | Health hazard | • Ventilate the workplace sufficiently.<br>• Observe the notes in the operating manual. |
| **Fire and explosion** | | | |
| Fire hazard | Entire system | Risk of injury | Estimate danger of fire as part of the operational hazard analysis. Implement appropriate fire protection measures in accordance with valid national regulations. |

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| | | | |
|---|---|---|---|
| | Compact dust extractor | Risk of injury | Keep a CO2 hand-held fire extinguisher ready (fire classification B).<br>• Keep a powder hand-held fire extinguisher ready (fire classification A,B,C). |

Table 1-4

The following overview of residual risks is a summary of the main potential threats to life and limb posed by the machine. Any additional precautions that can be taken by the machine owner to reduce residual risks are specified in the overview of residual risks.

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# Chapter 2 Product Overview

## 2.1 The most important assemblies

The laser cutting machine is composed of several essential components that work together to ensure high efficiency and precision during the cutting process. Below is a detailed overview of the key parts and their functions:

### 2.1.1 Machine Bed

The machine bed serves as the foundational structure of the laser cutting machine, providing a stable framework. It is typically made from heavy-duty steel to ensure stability and resistance to vibration. The precision of the machine bed directly affects cutting accuracy. It generally features high rigidity and excellent damping performance. By adjusting the leveling feet, the machine bed can be aligned to maintain balance, preventing uneven surfaces from affecting the cutting quality.

### 2.1.2 Crossbeam

The crossbeam is a critical component responsible for moving and supporting the cutting head as it moves across the worktable. It is usually made from lightweight, high-strength materials like aerospace-grade aluminum, ensuring smooth and efficient movement. The strength and weight of the crossbeam are crucial for achieving high-speed, high-precision cutting operations.

### 2.1.3 Control System

The control system acts as the “brain” of the laser cutting machine, executing numerical control instructions and coordinating the laser source, cutting head, and movement axes. Common control system software optimizes cutting paths, sets cutting parameters, and provides remote monitoring and diagnostics. The system’s stability and user-friendly interface directly impact the operator’s experience and the machine’s processing efficiency.

### 2.1.4 Laser Source

The laser source is the core component of the cutting machine, responsible for generating a high-energy laser beam. Fiber lasers are known for their high electro-optical conversion efficiency, making them suitable for cutting

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metal materials with low maintenance costs. The power and wavelength of the laser determine its cutting capabilities and the range of materials it can handle.

### 2.1.5 Cutting Head

The cutting head is the component that emits and focuses the laser beam. Positioned above the work area, it directs the laser onto the material surface for cutting. The cutting head is equipped with focusing lenses and sensors to ensure real-time adjustment of focal distance and height during the cutting process, achieving optimal cutting quality. It can also be fitted with an automatic following system to adapt to the cutting requirements of materials with varying thicknesses.

### 2.1.6 Cooling System

The cooling system is essential for regulating the temperature of the laser source, cutting head, and other high-temperature components to ensure stable operation. The most common cooling method is a water-cooling system. The water chiller circulates cold water to absorb the heat generated during operation, maintaining the laser source temperature within a specified range and preventing overheating. The quality of the cooling water must be strictly controlled to avoid scaling and impurities that could impact cooling efficiency.

### 2.1.7 Gas System

The gas system provides auxiliary gases for the cutting process, such as oxygen, nitrogen, or compressed air. Different gases have different effects on the cutting process: oxygen is typically used for carbon steel cutting, while nitrogen is used for stainless steel and aluminum alloys. Air can also be used for cutting stainless steel, carbon steel, and aluminum sheets at a lower cost. The purity and pressure of the gas directly affect cutting speed and edge quality, so the gas system is often equipped with filters to ensure gas purity.

### 2.1.8 Water Circulation System

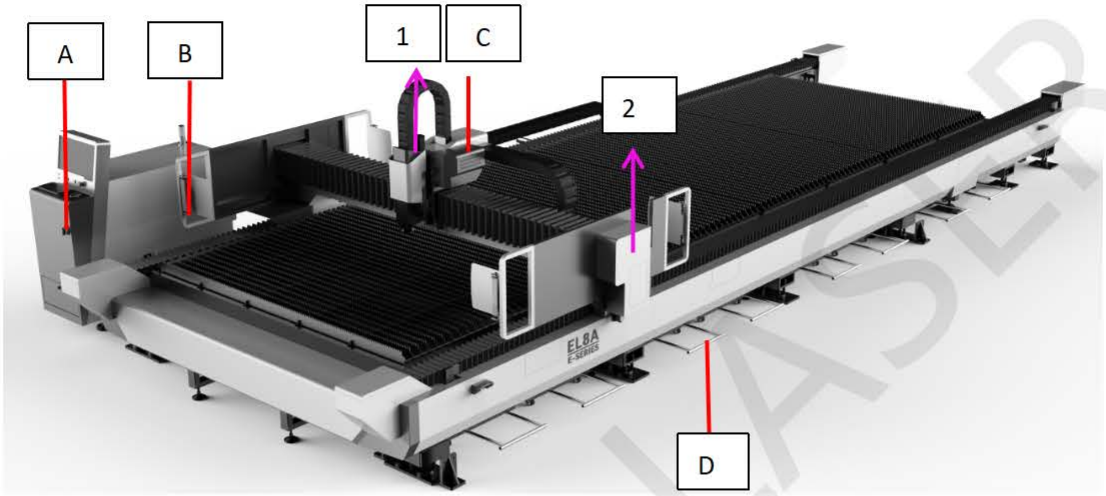
The water circulation system is connected to the cooling system, providing cooling water for the laser source, cutting head, and other critical components. The water system is divided into a primary loop and an external optics loop. The primary loop cools the laser generator, while the external optics loop is responsible for cooling optical components like fiber connectors and lenses. Proper control of water quality and temperature is crucial to prevent equipment damage and ensure stable operation.

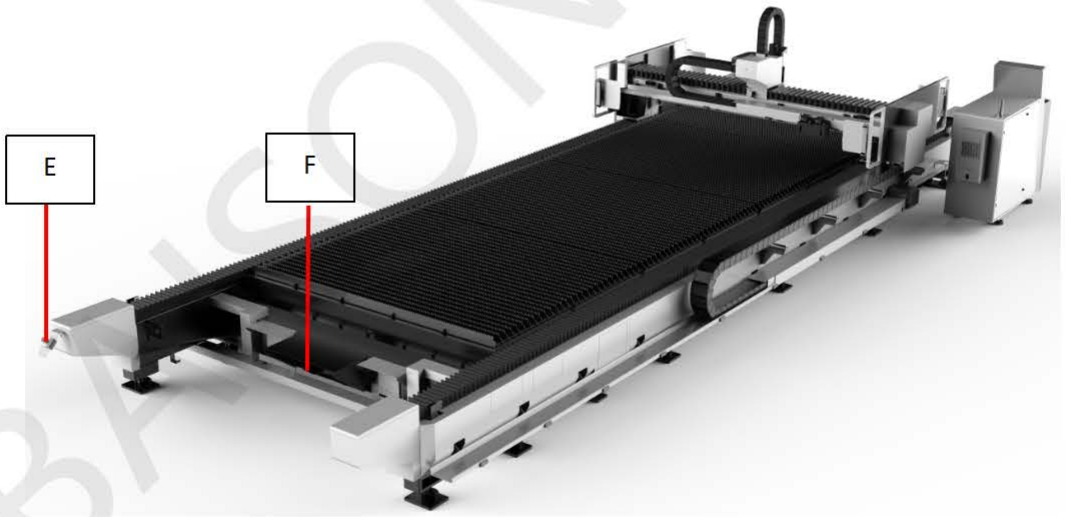
### 2.1.9 Dust Extraction System

During the cutting process, significant amounts of dust, smoke, and metal debris are generated. The dust extraction system is responsible for removing these by-products to maintain a clean working environment and

prevent dust from affecting the equipment. Dust collectors or fume extractors are commonly used to capture and filter the particles produced during cutting, ensuring a clean workshop atmosphere and protecting both workers and equipment.

## 2.2 Product Display

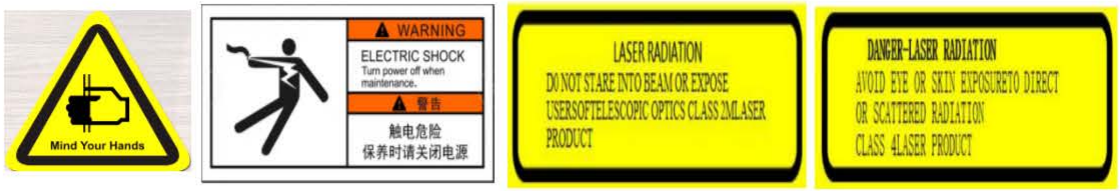



| | | | |
|---|---|---|---|
| A | Operation Console / Electrical Cabinet | B | Safety Grating |
| C | Proportional Valve | D | Waste hopper |
| E | Emergency Stop Button | F | Exhaust Port |

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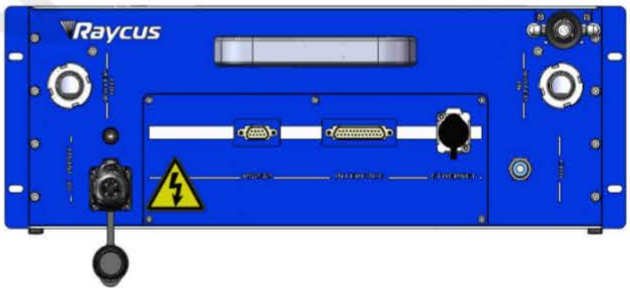
### Danger signs on the machine

| | |
|---|---|
| 1 |  |
| 2 |  |

Note: The product illustrations shown in this manual are for reference purposes only. The actual appearance of the product may vary due to model updates or design optimizations. Please refer to the actual product for specifics. The functionality and operation methods remain unaffected by appearance changes.

## 2.2.1 Laser Source




| | |
|---|---|
| LASER | The laser is emitting light normally. When the green light is on, it means that the laser is emitting light. |
| POWER | Control system power indicator light, when the red light is on, it means that the laser is powered on. |
| ALARM | Alarm indication, when the yellow light is on, it means that the machine has an alarm. |
| WATER OUT | Pipe connectors, the laser chiller water outlet |
| AC INPUT | The socket for supply source input. Please use only the enclosed power cord provided by Raycus. |
| POWER | An air switch that controls the on/off switching of AC power. |
| CDA | Dry air inlet. |
| ETHERNET | Ethernet interface. It can provide remote control and storage alarm information for the laser. |
| CTRL-INTERFACE | Control interface, CTRL-INTERFACE interface (DB-25), multi-function multiplex interface, |

| | |
|---|---|
| | users can set control mode, input analog voltage signal, modulate 24V signal, and it is also an alarm signal output interface. |
| WATER IN | Pipe connectors, the laser chiller water inlet |
| RS232 | Provide remote control and fault information storage of the laser |

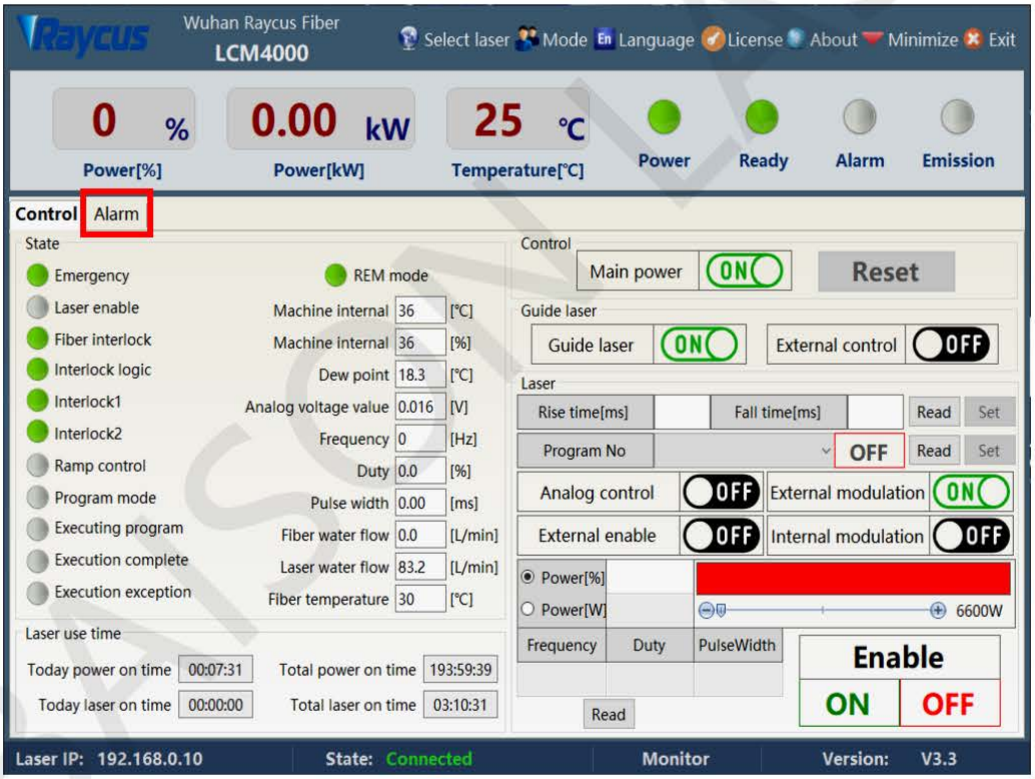
## Raycus Laser Monitoring

The laser monitoring function can only be used after connecting to a computer, with connection options including the 232 serial port mode and the Ethernet connection mode.

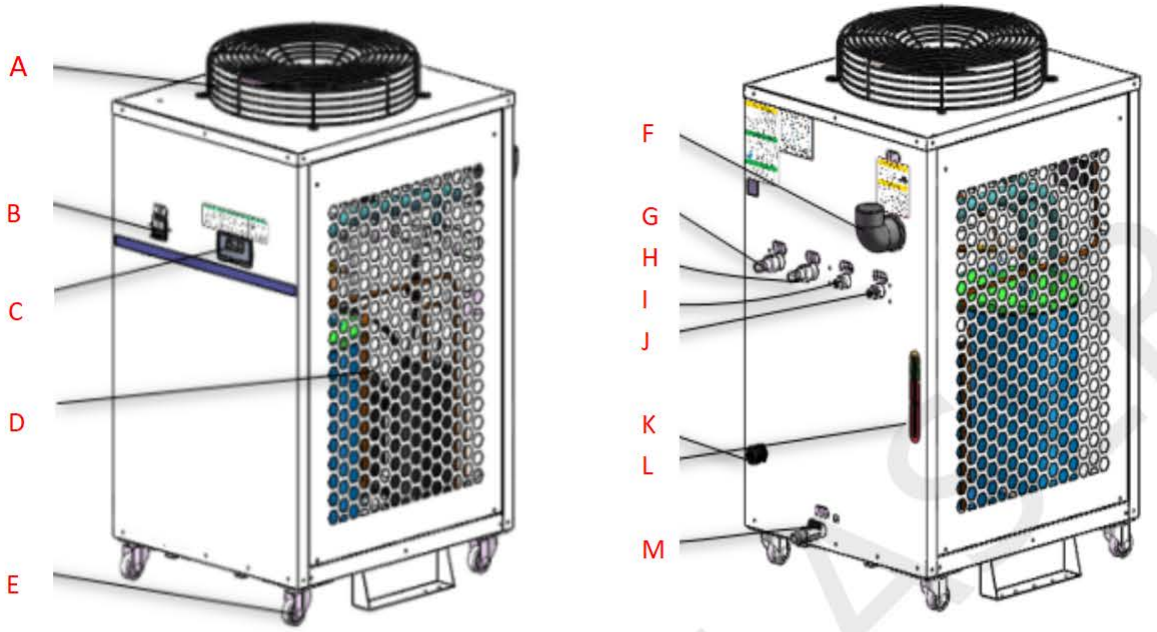
If using the 232 serial port mode, the corresponding COM port must be selected for monitoring.

If using the Ethernet connection mode, the connection must be established via IP address on the computer.

The IP address of the laser is 192.168.0.178, and the IP address of the computer is 192.168.0.180. After confirming the connection, click “Login” to enter the monitoring system.



## 2.2.2 Chiller




| | |
|---|---|
| A | Exhaust Port |
| B | Circuit Breaker Leak Compensator |
| C | Controller |
| D | Air Intake |
| E | Caster |
| F | Water Filling Port |
| G | Low-Temperature Cooling Water Outlet |
| H | Low-Temperature Cooling Water Inlet |
| I | High-Temperature Cooling Water Outlet |
| J | High-Temperature Cooling Water Inlet |
| K | Power Cord Outlet |
| L | Liquid Level Indicator |
| M | Drain Outlet |

### Hanli Temperature Control Panel

The set temperature for normal temperature water = [low-temperature water set temperature] + [F01 normal water temperature difference]. Modifications require changing the manufacturer parameter [F01 normal water temperature difference].

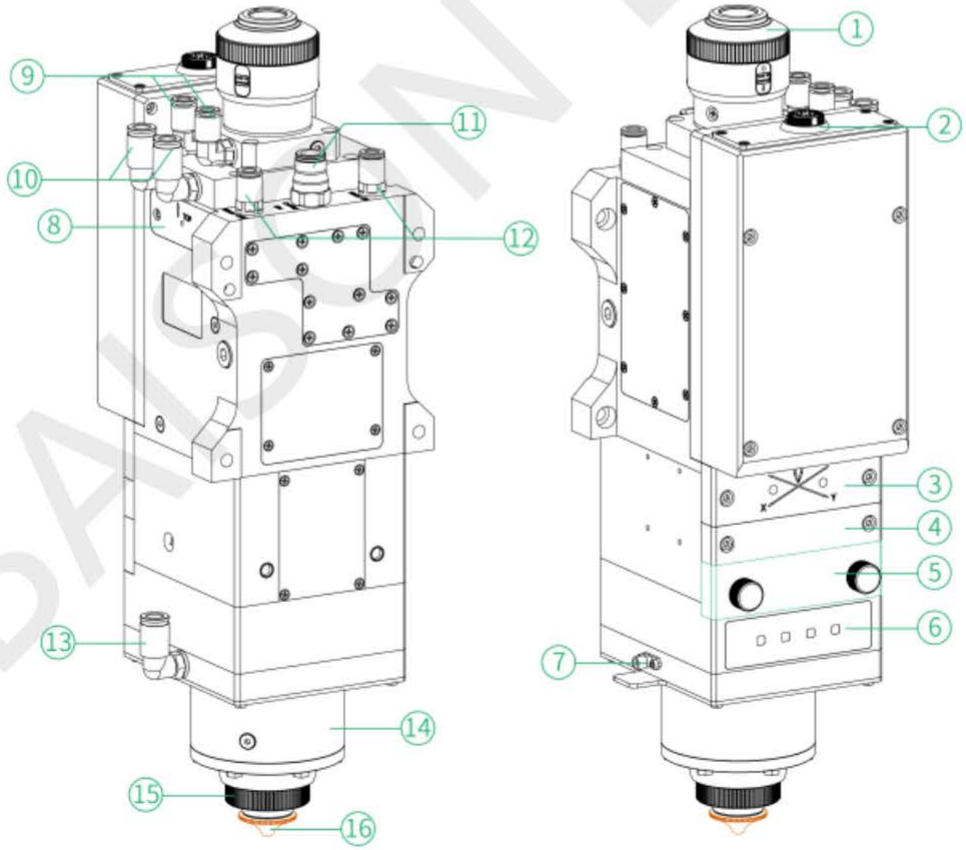
On the temperature display interface, press the <▲> and <▼> keys simultaneously for 5 seconds to enter the manufacturer parameter setting mode. Factory-set parameters are generally not adjusted; if adjustments are needed, please seek approval from the chiller manufacturer.

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During parameter setting, you can modify the parameter values using the <▲> or <▼> keys. If no key is pressed for 5 seconds, or if <▲>+<▼> is pressed, the system will return to the parameter selection mode and save the changes (the display window will flash the parameter value).




## 2.2.3 Cutting head




1 Fiber Interface

2 I/O interface

3 Focus Module (XY Alignment)

4 Middle Cover Glass Assy
5 Bottom Cover Glass Assy
6 Status display (LED)
7 Preamplifier Interface
8 Upper protective lens (collimation dual protection)
9 &10 &12 Water Cooling Interface (Φ6)
11 Cutting Gas Interface (Φ10)
13 Nozzle Cooling Gas Interface (Φ8)
14 TRA (Nozzle Assy)
15 Retainer Ring
16 Ceramic Body & Nozzle

## 2.3 Product Specification

| Model | EL8A Fiber Laser Cutting Machine | |
|---|---|---|
| Laser Power | Raycus | 6000W |
| Central wavelength | 1080±5(nm) | |
| Processing Area | X/Y Axis | 3350/13100mm |
| Accuracy | Z Axis | 200mm |
| | X/Y Axis Positioning Accuracy | ±0.03mm |
| | X/Y Axis Repeat Positioning Accuracy | ±0.03mm |
| Ambient Temperature | Permissible Ambient Temperature | +10 to 35 °C |
| Dimensions | Length | 11100mm |
| | Width | 3768mm |
| | Height | 2348mm |
| Electrical Parameters | Machine Power Supply/Voltage/Frequency | AC380V/50/60HZ |
| | Power Protection Level | IP54 |
| Cutting Range | Oxygen for carbon steel | 1-22mm |
| | Nitrogen for stainless steel | 1-18mm |
| Note: The parameters marked in red in the table are the proofing thicknesses, which are greatly influenced by various factors in actual processing and are not recommended for mass production. It is suggested to use a higher power laser. | | |

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# Chapter 3 Installation

## 3.1 Unpacking Instructions

Packaging Removal: If the equipment is packaged in a wooden crate, follow the external instructions to remove it carefully to avoid damaging the machine. For equipment wrapped in protective film, avoid using sharp tools that may tear the packaging and scratch the equipment surface or damage the electrical installation's protective conduits. The company will not be responsible for any damage caused by improper handling.

### Inspection Items

Confirm the Equipment Model: After unpacking, immediately verify that the equipment model matches the purchased product to ensure it meets the order specifications.

Check for Shipping Damage: Conduct a thorough inspection of the equipment to check for any damages or abnormalities that occurred during shipping.

Verify Completeness of Accessories: Verify that all parts and accessories listed in the accompanying documentation are present and undamaged. Pay special attention to the integrity and functional status of key components.

Handling Abnormalities: If discrepancies in model, missing accessories, or transport damage are discovered, promptly contact the company's after-sales service department or distributor to address and resolve the issues.

### Other Considerations

Photograph and archive: It is recommended to take photos during unpacking to serve as documentation for addressing any future issues.

It is advisable to complete the aforementioned inspections promptly after unpacking to ensure the equipment's integrity and functionality.

## 3.2 Installation Instructions

Proper installation of the laser cutting machine is critical to ensuring the equipment functions correctly. Follow these instructions carefully.

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## 3.2.1 Site Requirements

- The floor should be level, with height differences within the laser cutting machine foundation area not exceeding ±10mm. It must withstand a weight of over 20 tons to avoid operational errors caused by foundation issues.
- Maintain at least 1.5 meters of space between the laser and the wall, and the distance between the chiller and the laser should not exceed 10 meters to ensure smooth connection of the cooling pipes.
- The laser cutting machine should be installed on a vibration-free floor and should not be used in conjunction with equipment that produces impact or vibration.
- Leave a minimum of 1.2 meters of operational space around the machine for daily maintenance and personnel access.
- Avoid installing the laser cutting machine in environments with flammable or explosive materials. Keep the machine away from open flames and welding operations.

## 3.2.2 Environmental Requirements

### a) Temperature Control

The ambient temperature should be maintained between 15°C and 30°C to prevent high or low temperatures from affecting the equipment's performance. If the temperature exceeds 35°C in the summer, it is recommended to use cooling measures to prevent the laser from overheating. In winter, if the temperature drops below 10°C, the water cooling system must be kept running, or antifreeze should be added to prevent the system from freezing and causing damage.

### b) Humidity Requirements

The relative humidity should be controlled between 30% and 70%. High humidity may cause electrical components to become damp, reducing the stability of the laser, while low humidity may cause static electricity, damaging electronic components.

### c) Cleanliness Requirements

The operating environment should be kept clean to prevent dust, oil mist, or other pollutants in the air. These particles can affect the cleanliness of optical components, reducing beam quality.

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Avoid installing the machine in areas where paint, solvents, or degreasing processes produce smoke, as this can absorb laser radiation and affect performance.

### d) Airflow Control

Avoid strong airflow or unidirectional winds around the equipment to prevent interference with the optical path system or machine tool movements. If the equipment is installed near windows, use curtains or blinds to prevent direct sunlight.

### e) Ventilation Requirements

The workshop should be equipped with a good ventilation system to ensure proper airflow but avoid direct airflow toward the laser cutting machine. Smoke and harmful gases generated during cutting should be promptly removed via the exhaust system to ensure the health of operators and the cleanliness of the equipment.

## 3.2.3 Electrical Requirements

Connect the power supply according to equipment requirements, with a voltage range of 380V ± 5%. A voltage regulator must be installed. Ensure proper grounding, with the grounding rod reaching a depth of 2 meters and the ground resistance being ≤ 0.5Ω to prevent static electricity from damaging the equipment and harming operators.

| | |
|---|---|
| **Power** | Three-phase five-wire system |
| **Voltage** | AC380V 50/60HZ |
| **Voltage fluctuation** | ±5% |
| **Wire material** | Copper |
| **Power cable size** | 12-14 square millimeters |
| **Voltage regulator range** | 100KVA |

## 3.2.4 Water Circuit Connection

The water system of this machine consists of two main parts:
Main Circuit: The main circuit provides cooling water for the laser generator to ensure its normal operation. In summer (when the temperature is above 35°C), the cooling water temperature for the main circuit should be set to 27°C; in winter (when the temperature is between 10°C and 20°C), the cooling water temperature should be set to 24°C. If the temperature is below 10°C, it is recommended to keep the chiller running to avoid freezing risks, or add designated antifreeze as needed.

External Optical Circuit: This circuit is used to cool key optical components such as QBH fiber connectors and collimating lenses. In summer (when the temperature is above 35°C), the cooling water temperature for the external optical circuit should be set to 30°C; in winter (when the temperature is between 10°C and 20°C), it should be set to 28°C.

The inlet and outlet water pipes of the chiller should be properly connected to ensure the cooling system operates correctly.

The inlet and outlet of the chiller should be color-coded to ensure the correct water flow direction.

## Filling

Open the inlet valve and fill the water tank. The water level should be 30mm-50mm below the top edge of the tank to prevent overflow. Once the water temperature reaches the set value, the chiller will automatically control the temperature, and the water pump inside the unit will continue to operate.

## Attention

Before the initial startup, make sure to open the vent valve of the water pump to release air; otherwise, the pump may be damaged. Adjust the valves in the water circuit system as needed, but avoid adjusting them during subsequent operations.

The laser can only be turned on when the water temperature and the pressure difference between the inlet and outlet meet the laser's requirements.

The chiller is used to cool the laser generator and cutting head, and only pure or distilled water should be used during operation. Since scale can form during the circulation of cooling water, affecting the chiller's operation, it is recommended to replace the water every 1 to 2 months to keep the cutting head and chiller pipes clean and free from contamination or blockages.

In areas where the temperature drops below 0°C, antifreeze must be added to the chiller during winter to prevent the cooling fluid from freezing, which could cause the pipes inside the laser and chiller to burst, damaging the seals.

## The antifreeze requirements are as follows:

- Excellent antifreeze performance: Ensure normal operation under low temperatures.
- Anti-corrosion and anti-rust performance: Effectively protect the internal components and pipelines from corrosion.
- Rubber seal compatibility: Does not dissolve or corrode rubber-sealed pipes.
- Moderate low-temperature viscosity: Ensures cooling circulation efficiency under low temperatures.
- Stable chemical properties: Prevents chemical changes during long-term use.

While ensuring antifreeze performance, it is recommended to choose antifreeze with lower concentration and shorten the usage cycle. Avoid mixing different models and brands of antifreeze to prevent adverse chemical reactions that may affect equipment performance.

## 3.2.5 Gas Requirements for Equipment

The cutting process of the laser cutting machine requires the use of different types of auxiliary gases, with the specific requirements as follows:

**a) Gas types:**
High-purity oxygen: Used for cutting carbon steel materials, it helps improve cutting speed and quality.

High-purity nitrogen: Used for cutting stainless steel, aluminum, and other metals, it effectively prevents oxidation and discoloration of the cutting edges.

Air: A low-cost alternative gas, suitable for non-high-precision cutting needs, typically used for rough processing of aluminum and carbon steel.

**b) Gas requirements:**
Gas cylinder or cylinder group supply: Gas cylinders or cylinder groups are one of the simplest methods of providing cutting gases. When using bottled oxygen or nitrogen, ensure the gas purity is above 99.7%, as insufficient purity will severely affect cutting performance.

Liquid oxygen supply: When using liquid oxygen as the cutting gas, the gas pressure should reach 1 MPa, and the purity should be 99.9%. The supply system must be connected to the storage tank via the liquid phase, and the vaporizer's gas supply capacity should be at least 50 cubic meters.

Liquid nitrogen supply: When using liquid nitrogen gas, the pressure should remain at 2-3 MPa, and the purity should reach 99.99%. Similarly, the system must be connected to the tank via the liquid phase, and the vaporizer's gas supply capacity should be at least 100 cubic meters.

**Note:**
When changing gas cylinders or cylinder groups, the gas flow will be interrupted.
The connecting pipes must be clean, ensuring they are free of oil, grease, or dust inside and out.
Do not use solvents to clean hoses.
Do not use lubricants when connecting hoses.
All maintenance work must be carried out under extremely clean conditions.
A pressure regulator must be installed at the gas source between the gas cylinder and the machine.

**Pressure regulator requirements:**

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Oxygen pressure regulator: The output pressure should reach 0.1-1.5 MPA.
Nitrogen pressure regulator: The output pressure should reach 2-3 MPA.
When the blowing function is activated in the cutting software, the pressure drop on the gas gauge must not be lower than 0.3-0.4 MPa. A large pressure drop will affect the cutting quality.



Oxygen pressure regulator



Nitrogen pressure regulator

(Images are for reference only)

**c) Air Cutting Gas Supply:**

For air cutting, a screw-type air compressor should be used with a pressure of 2.5 MPa and an exhaust volume of at least 2.0 cubic meters per minute. To ensure the quality of the gas, a minimum of three to five stages of filtration equipment should be installed (the number of filters can be increased depending on the environmental conditions). Additionally, an air dryer with a processing capacity of 3.8 cubic meters per minute should be used, along with an adsorption dryer and a 1.2 cubic meter air storage tank.



Screw-type Air Compressor



Air Dryer



Adsorption Dryer

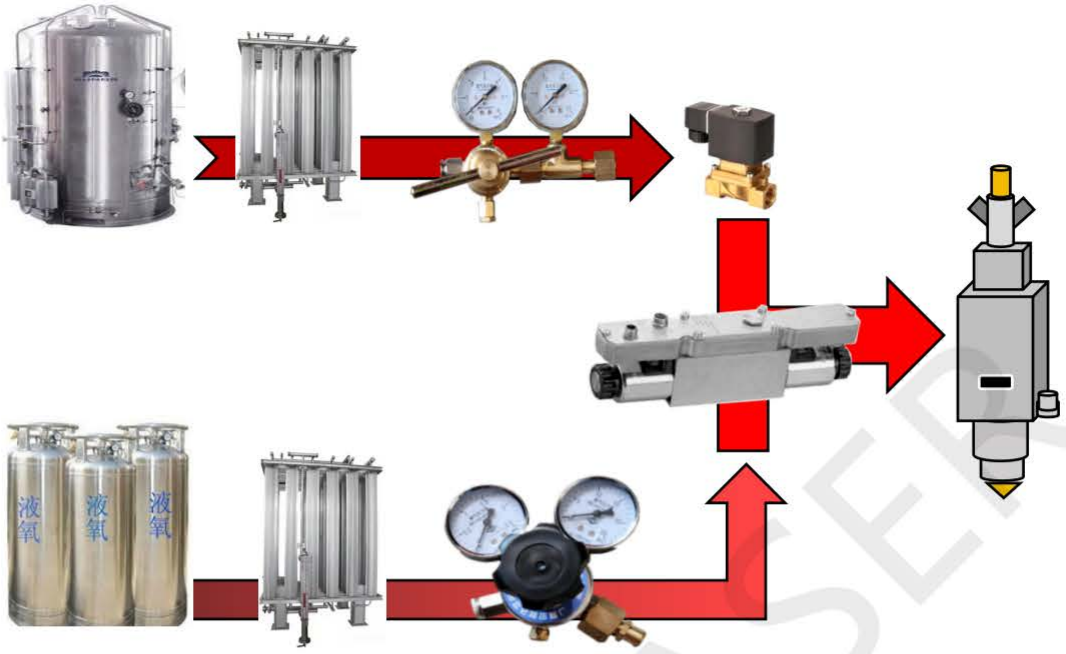


Air Storage Tank

(Images are for reference only)

**d) Gas Circuit Connection Check**

Ensure that the gas pipes are firmly connected to avoid leaks. Before each use, inspect the gas pressure gauge to ensure sufficient gas supply.




(Images are for reference only)

Note:

• During the operation of the air compressor, the compressed air inevitably contains oil and water (moisture in the air). If the filtration is inadequate, it may cause serious damage to the gas circuit system components and the cutting head. The user is solely responsible for any equipment damage caused by poor filtration.

• The gas consumption is closely related to the thickness of the cutting material. Adjust the gas usage as needed according to the actual requirements.

• When using gases, pay special attention to their flammability and explosiveness. Ensure that they are kept away from open flames and welding operations, and operate safely.

## 3.2.6 Cleaning Supplies Requirements

**a) Anhydrous Ethanol**

Anhydrous ethanol or isopropanol with a purity of ≥99.7%, used for cleaning and wiping the protective lenses of the laser cutting machine, ensuring the cleanliness of optical components.

**b) Masking Tape and Transparent Adhesive Tape**

Two rolls of masking tape (50mm width) and transparent adhesive tape, primarily used to seal the cutting head to prevent dust and contaminants from entering. Additionally, these tapes can be used to observe the center point adjustment of the cutting head offset and the center point of the copper nozzle, ensuring cutting accuracy.

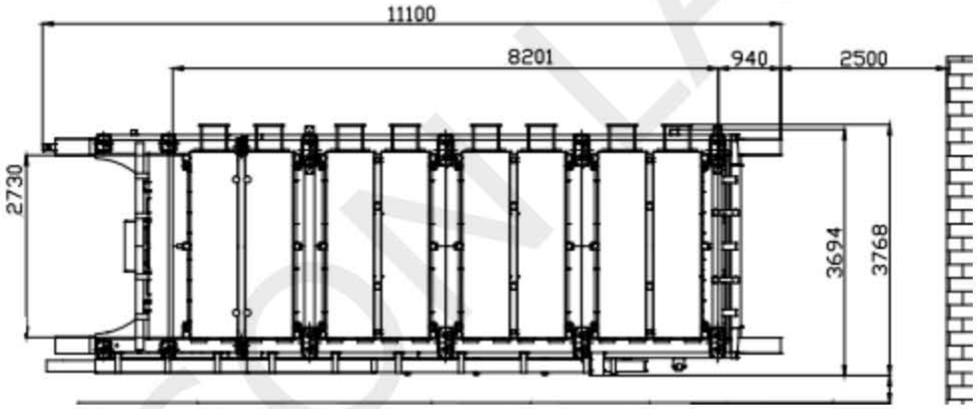
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(Images are for reference only)

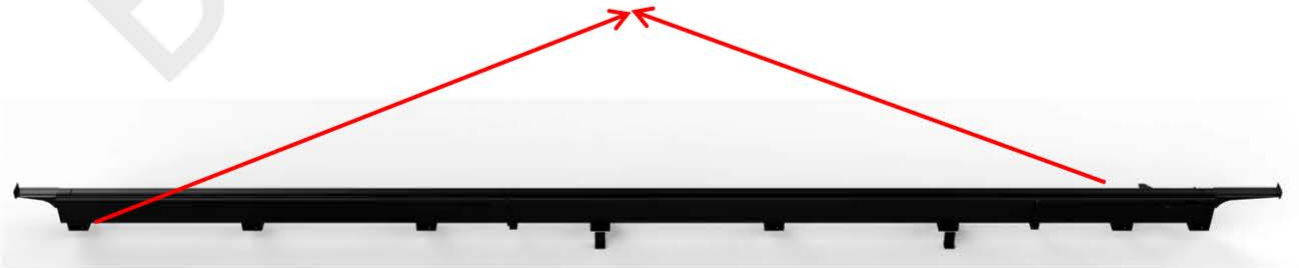
## 3.2.7 Machine Installation

### Footprint Dimensions




### Lifting Method

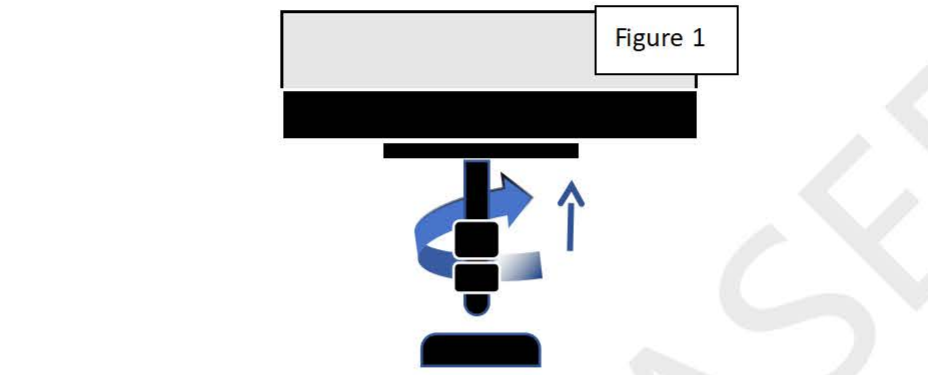
The segmented bed body transportation can be unloaded directly by using a forklift with a capacity of 5-ton or above.For lifting, use a 5-ton crane with the equipment having lifting rings. Check if the lifting rings are installed before lifting, and secure the lifting rings for loading and unloading.



### Screw and Foot Cup Installation

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Put on protective gloves and first remove the bolt, then install a nut, and then rotate the bolt clockwise to install it back onto the base, placing the bottom of the bolt into the foot cup (ignore this step if it comes with 2 nuts).
Rotate the bolt in the middle of the equipment to the highest position without touching the foot cup, leaving the front and back foot cups of the equipment in contact with the ground (as shown in Figure 2, note that the distance between nuts should be 12-14mm). After the equipment is properly positioned, you can place the foot cup under the bolt.

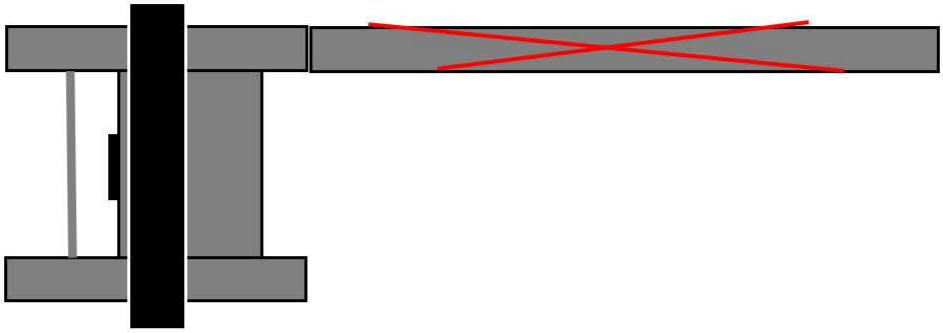



## Machine Leveling Adjustment

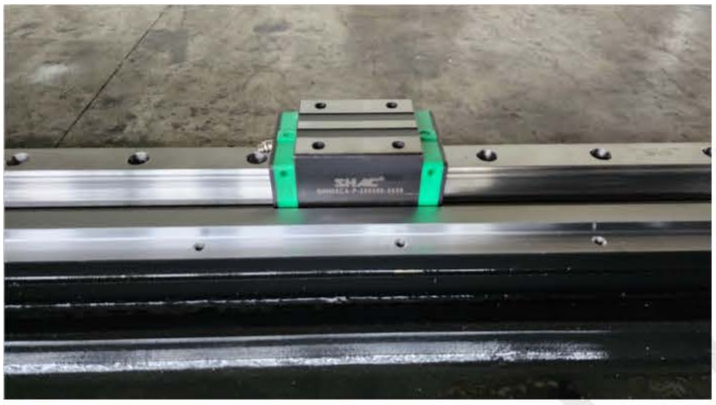
1. Place the bed frame with crossbeam onto the pre-measured and designated space. Ensure that the front and rear support screws are at the same height, and rotate the remaining screws to their highest position. Use a spirit level to level the machine bed along the X-axis, Y1, and Y2 axes.
Place the spirit level on the guide rail. Observe the direction of the bubble shift: if it shifts to the left, lower the left-side screw or raise the right-side screw to level the bed.

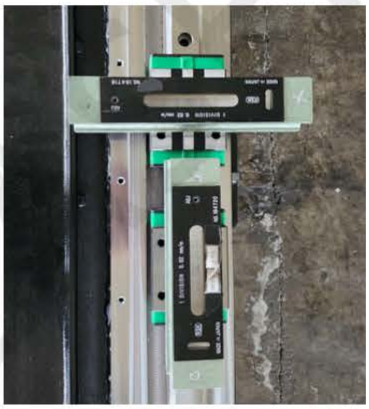
2. Once the rear-end bed with the crossbeam is leveled, hoist the Y1-axis side bed toward the interface of the rear bed. After lowering, install the slider to make close contact with the rear bed. Adjust the screw height to match the rear-end bed height.

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3. Clean the guide rail with a clean cloth. Place the spirit level on the slider. Adjust the height of the support screws based on the bubble direction. Slide the slider along the bed to check levelness at multiple positions.





4. Hoist the right-side Y2-axis bed close to the rear-end bed, and align using the same method as for Y1. Level the bed and tighten the fixing screws. Push the crossbeam onto the installed Y1 and Y2 bed frames. Check the precision of the X-axis. Set up a granite reference surface and use a dial indicator to measure its parallelism with the Y1-axis guide rail. Use the dial indicator along the transverse side of the granite to check X-axis precision accordingly.



5. Use the first machine bed section as a height reference to align the remaining sections. After bringing two sections together, pre-tighten the screws, use a dial indicator to check the rail alignment precision, and adjust the

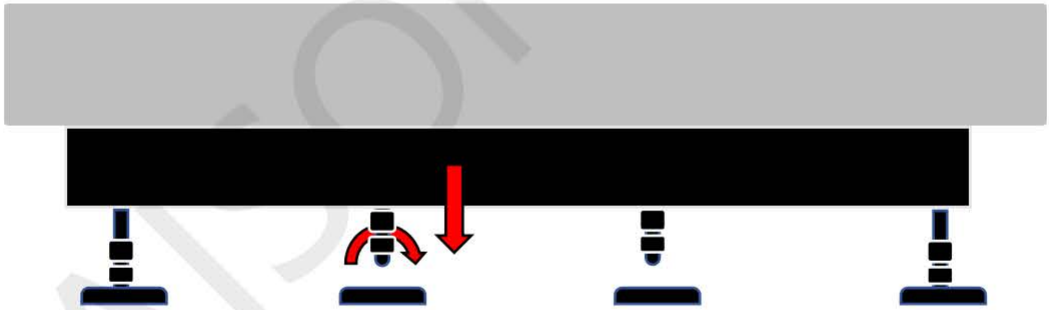
bed angle if needed. Once complete, tighten the locknuts and fix the leveling foot to prevent loosening.





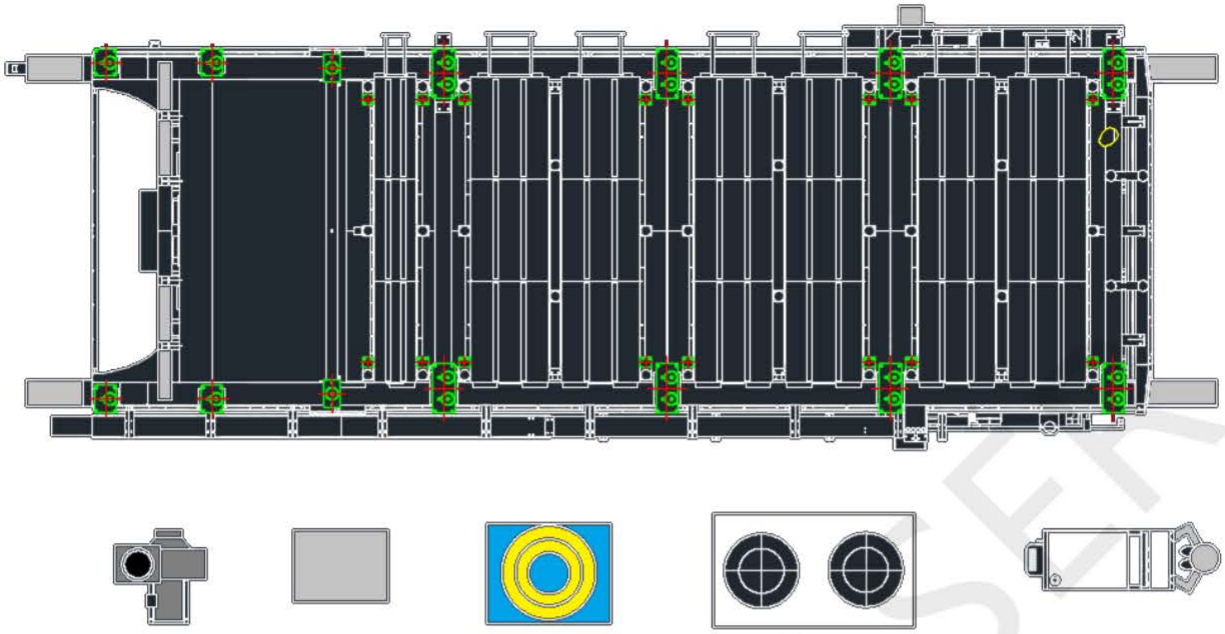
## 3.2.8 Adjusting the Leveling Screws

After the bed has been leveled, wear protective gloves and screw down all suspended leveling screws until they contact the leveling feet. Use an adjustable wrench (⩾300mm) to pre-tighten the nut that contacts the leveling foot, then hold this nut in place. Use another adjustable wrench to tighten the upper nut. After tightening each leveling foot nut, twist to check if the foot is loose.



## 3.2.9 Installation of the Laser, Cutting Head, and Chiller

Remove the laser and cutting head from the box, place the laser in the position indicated in the diagram below, and place the chiller behind the equipment to connect the water hoses.

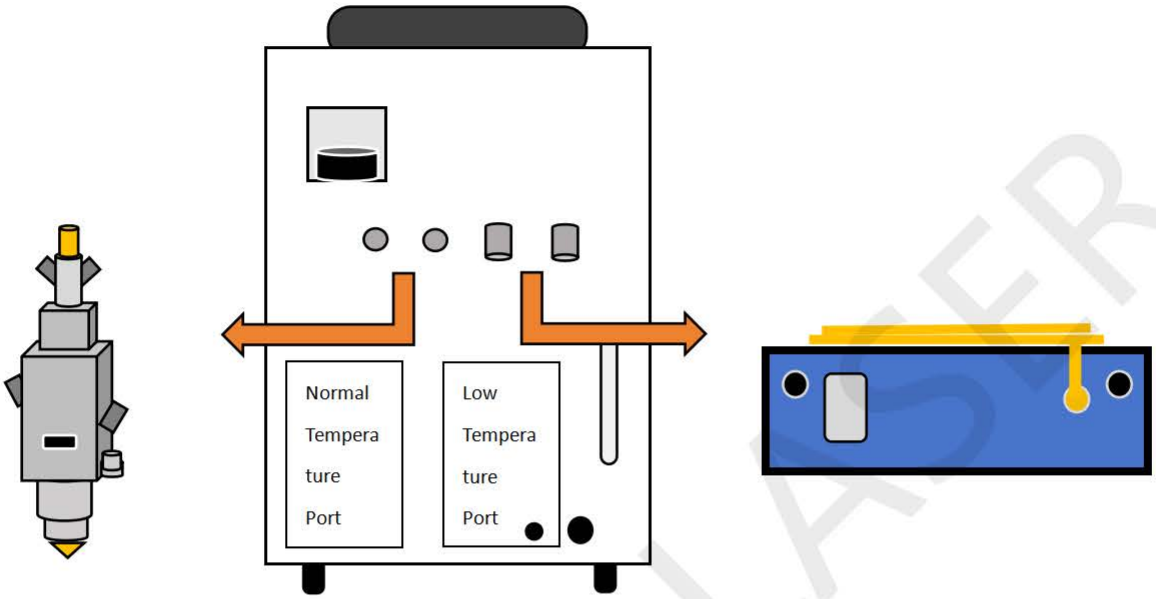


Using a small flat-head screwdriver, remove the cover of the Z-axis cable carrier. Install the cutting head in the corresponding position on the sliding platform (the yellow optical fiber cable must not be folded, stepped on, or bent with a radius smaller than 15 cm). Place the optical fiber into the Z-axis cable carrier, and connect the water pipes, air pipes, signal cables, and power cables to the cutting head. After placing everything inside the cable carrier, close the cover.





Normal temperature end: connect to the water pipe leading from the cutting head
Low-temperature end: the chiller is connected to the laser.
The water pipes of the chiller should be connected with IN corresponding to OUT. Lock the water pipe of the chiller with a throat band, and check whether the water flow switch is open after power on, and whether each water pipe will leak.




## 3.2.10 Electrical Connections

Machine power requirements: 380V—400V, 50Hz, with a mandatory voltage stabilizer.
① Verify that the main power and individual circuit breakers operate sensitively.
② Check that equipment power wiring is correct.
③ Ensure the main and sub-power circuit breakers (e.g., for main machine, laser, air compressor) meet the capacity specified in the schematic.
④ Ensure the gauge of power, ground, and neutral wires is not less than specified in the power wiring diagram.
⑤ Confirm the ground wire of the power cable is securely connected.
⑥ Inspect all high-power terminals, especially the input and output points of the power transformer, to confirm secure and reliable connections of all plugs and sockets.

**Warning:** The ground wire must be securely grounded. Failure to do so can lead to signal interference within the machine control cabinet and presents an electrical hazard in the event of a leakage.

Connect nitrogen/air hose, oxygen hose, and cooling air hose to the pressure gauge. Ensure pressures are set as follows: nitrogen at 2.0-3.0 MPa, oxygen at 0.8-1 MPa, and cooling air at 0.5 MPa.

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# Chapter 4 Commissioning Instructions

## 4.1 First Startup and Commissioning Steps

**a) Factory Power Supply Inspection**

Check the output voltage of the factory's main power switch to ensure the correct connection of the neutral (N) line. Wear insulated gloves, turn on the circuit breaker, and use a voltage tester to measure the voltage between the N line and live wires, as well as between the live wires. Each of the three-phase live wires must be measured individually to ensure no phase is missing. The input voltage of the voltage stabilizer should be 220V (between N line and live wire) and 380V (between live wires), ensuring that the N line is properly connected.



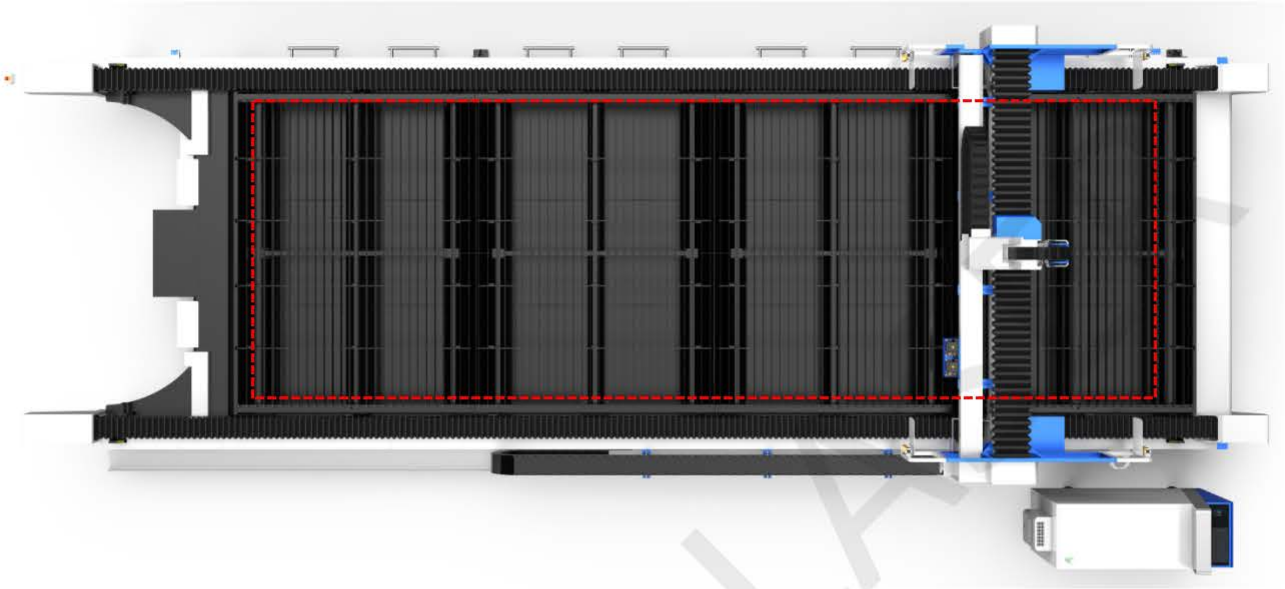

**b) Equipment Installation Check**

After the equipment installation is complete, inspect the electrical cabinet to ensure that all circuit breakers are switched on.



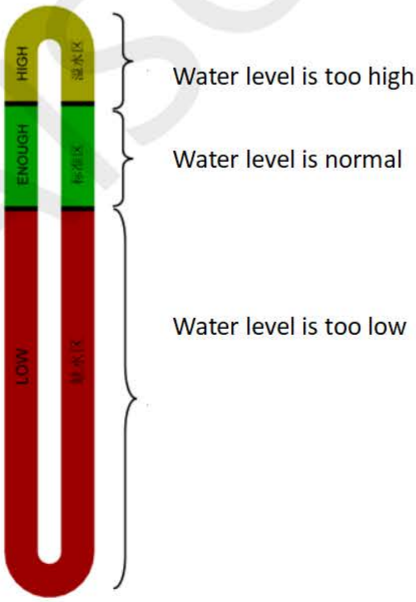
**c) Clearing the Movement Range**

Remove any foreign objects from the movement range of the equipment. Pay special attention to the red-marked areas, which is the rear equipment Y-direction movement range. Ensure there are no obstacles in these areas to prevent collisions.



**d) Chiller Inspection**

Check the water level in the chiller to ensure it is sufficient. During the first fill and startup, the chiller may trigger an alarm as the pipes and laser unit need to be fully filled with purified water. Turn off the chiller and refill it with water to the standard level. Check for any leaks at the water connection pipes; if there is a leak, immediately turn off the power and retighten the water connection hoses.



**e) Cooling Preparation**

The chiller operates for 5-15 minutes until the "controller" temperature display reaches 25-28°C, waiting for the dew point of the laser to decrease.

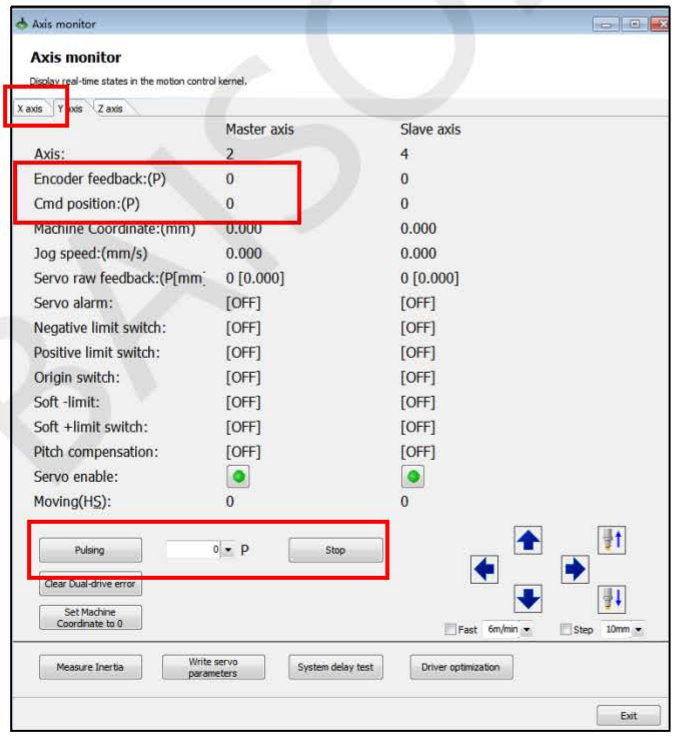
**f) Power On the Equipment**

Release the emergency stop button and start the computer. Once the system is up and running, press the servo button to activate the system.



**g) Launch Cutting Software**

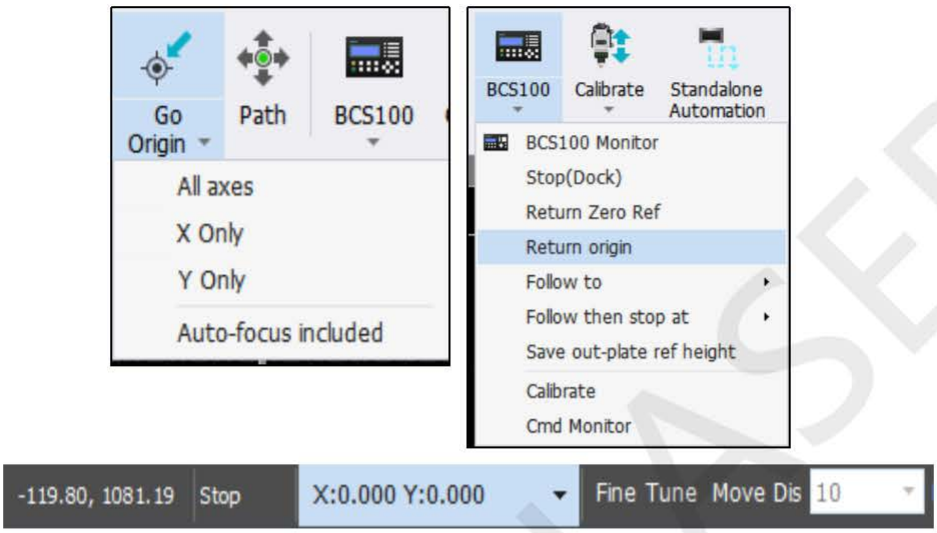
Open the cutting software “Cypcut”, manually jog each axis step by step, and check if the limit switches are functioning correctly. Ensure the machine is operating in normal working conditions.





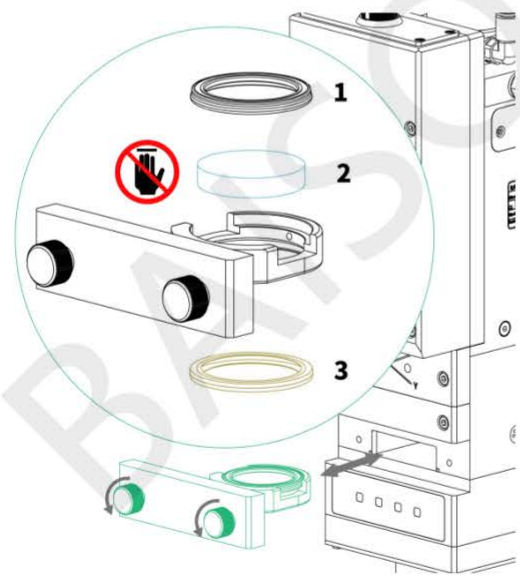

**h) Origin Setup**

After checking the origin and limit switches, you can proceed the return to origin action. When debugging for the first time, please perform a single-axis return to origin test first. Perform the single-axis home return operation for the X and Y axes in sequence. For the Z-axis, perform the home return operation within the "Bcs100" control system.




**i) Laser head inspection**

Check and clean the protective lens of laser cutting head , pay attention to whether there is a lot of smoke and dust in the surrounding environment, and avoid removing the protective lens in a smoky and dusty environment.

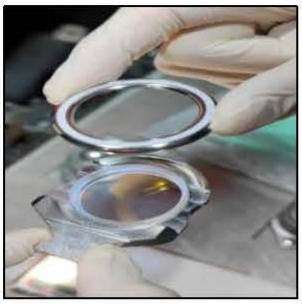
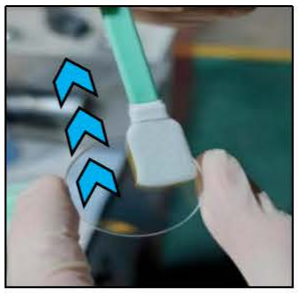



Step 1: Open the lower protective lens door

Step 2: Remove the lower protective lens holder

Step 3: Close the protective lens door

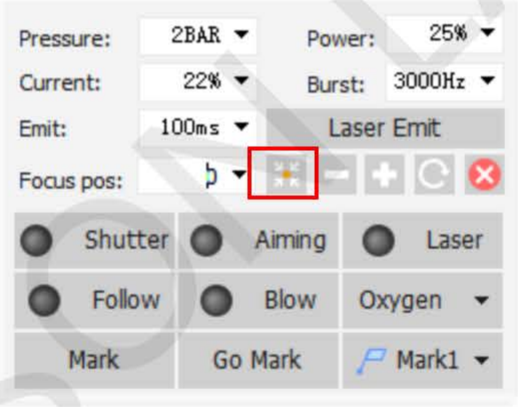
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| | | | |
|---|---|---|---|
|  |  |  |  |
| Wear clean gloves to pick up the protective lens ring. | Spray a little alcohol on the dust-free cotton swab | Pop out the excess alcohol on the dust-free cotton swab. Wipe the cotton swab in the same direction. | Wipe clean under strong light and then put it back into the lens holder. |

Step 4: Cover the pressure ring. Put it into the laser cutting head and close the protective lens door.

**j) Inspection of cutting head copper nozzle alignment**

Check whether the focus in the lower right corner of the software is 0. If it is not 0, change it to 0 and click to zoom.Ensure that the pulse power and duty cycle are appropriate, adjust them to the suitable parameters according to the image below.




Click on the handheld box shutter and check that the software shutter is off. Apply transparent glue to the bottom of the copper nozzle.

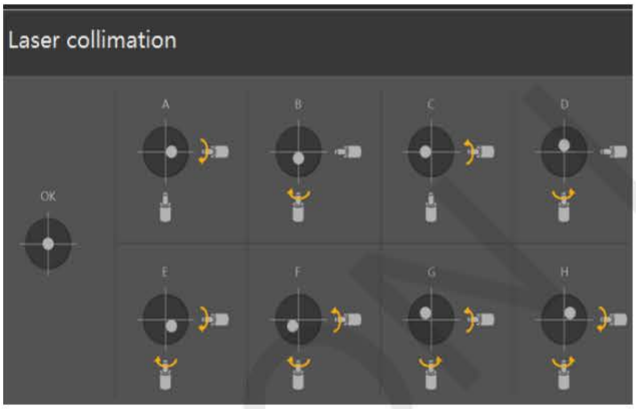
Click the shutter, click the laser point shooting software shutter , the light will light up

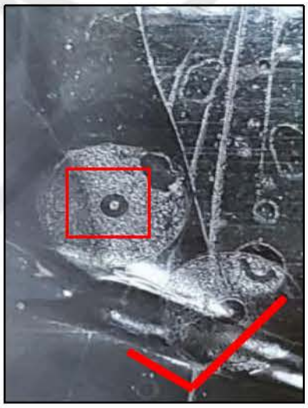
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Note: Some pictures may differ in content, please refer to the actual product

Take out the tape and observe the direction of the offset, adjust the diagram according to the light path centering, and adjust the center point to be centered with the outer ring.

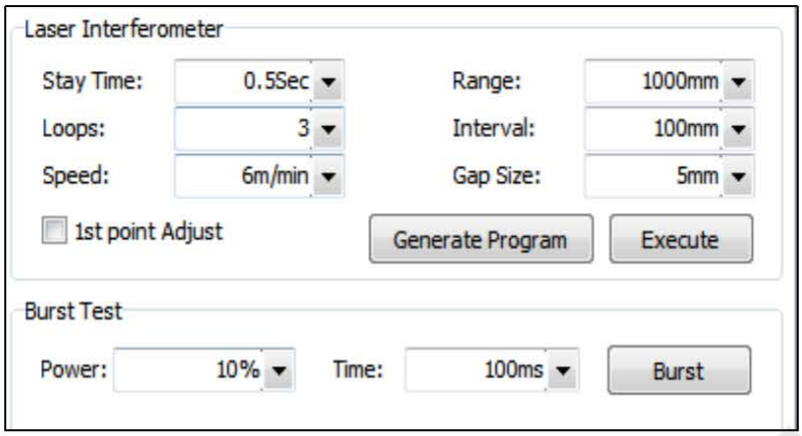


Note: When clicking on the laser, do not put your hands under the copper mouth to prevent burns. Do not place plates under the copper nozzle

**k) Check the Cutting Head with Photo Paper**

- Remove the copper nozzle and place thermal paper at a height of 150mm to 200mm from the ceramic ring of the cutting head.
- Set the air pressure to 0.3Bar and click the manual control box to activate the air blow. Click on the interferometer to set 100% power and duty cycle (with an output power of 3000W).
- Use the handheld controller to press the air blow button, confirm the detection position, and press the laser button to start the test.
- Inspect the thermal paper for any black spots. If a black spot appears in the white area (as shown in Figure 1), this may indicate that the lens is either dusty or damaged.
- Replace the protective lens and retest. By observing whether the position of the black spot changes, you can determine if the internal lens is contaminated.



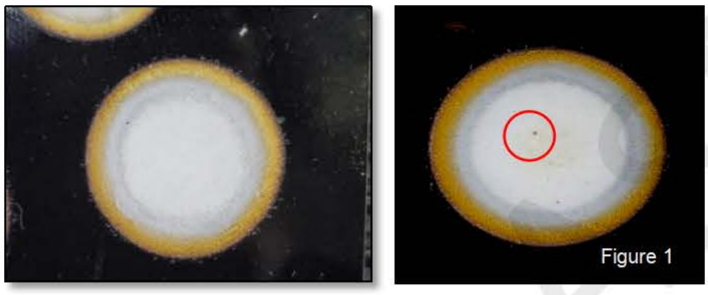



Figure 1

**I) Find focus adjustment**

**Focus position definition**

The distance from the focus to the upper surface of the cutting workpiece. The focus position above the workpiece is generally called positive focus, and the focus position below the workpiece is generally called negative focus.

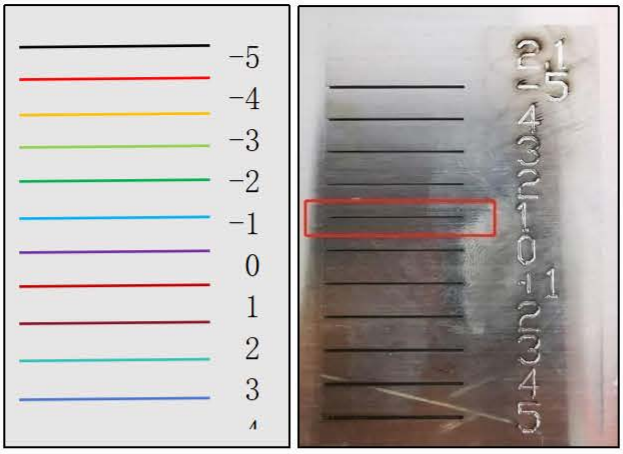
**Focus position meaning**

Changing the focus position means changing the spot size on the surface and inside of the plate. The focal length becomes larger, the spot becomes thicker, and the slit becomes wider and wider, which in turn affects the heating area, slit size and slag discharge ability.

**How to find zero focus**

Use 1mm stainless steel to find zero focus, and place the stainless steel under the cutting head. Click on the cutting head and select focus automatic test. Enter the focus range and adjust the length, spacing, and frame parameters of the line segment. Click to generate drawings, process corrections, and set various process parameters. Click to execute the cut line segment and observe which line segment corresponding to the focus is the thinnest, which is zero focus.

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**m) Capacitance calibration**

**Manual Calibration Procedure:**

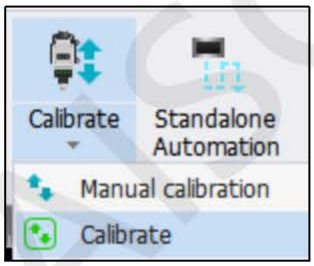
Move the cutting head above the plate, move the cutting head close to the plate about 5mm away.

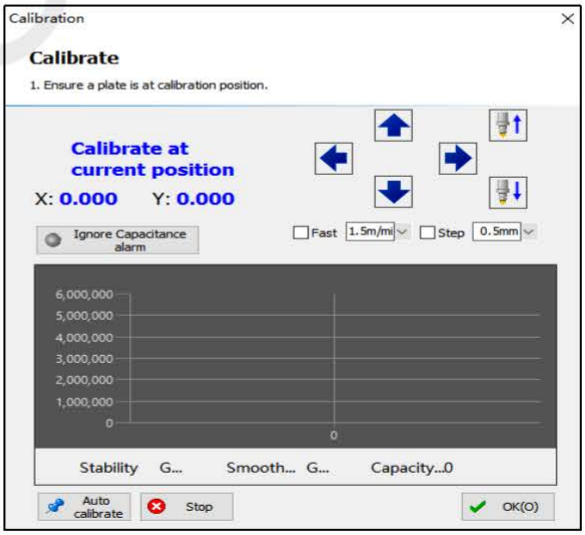
Click on "BCS100" to enter the calibration dialog box and click on capacitive calibration, which takes about 20 seconds to complete.




**Automatic Calibration Procedure:**

Move the cutting head above the plate and click on "Calibrate". Click to start auto calibration.




If calibration fails, refer to common troubleshooting issues:

| Indicator | Explanation |
|---|---|
| Stability | The difference in data collected during the descent of 5mm to touch the plate and the ascent of 5mm after touching the plate. The larger the difference, the worse the stability.If the stability is poor, it may be due to significant |

| | |
|---|---|
| | vibration or strong external interference, necessitating a re-calibration. |
| Smoothness | Refers to the smoothness of the curve.<br>If the smoothness is poor, it indicates that the curve is not smooth, has fluctuations, or has burrs, and re-calibration is needed. |

**n) Check gas pressure**

The oxygen pressure gauge should be adjusted to an output pressure of 0.8-1.0 MPa, and the nitrogen pressure gauge should be adjusted to an output pressure of 2.0-3.0 MPa.

Connect the gas meter to switch the gas type in the software, and click on the hand-operated box blowing/software blowing. Make sure there is gas coming out from the copper nozzle position of the cutting head.







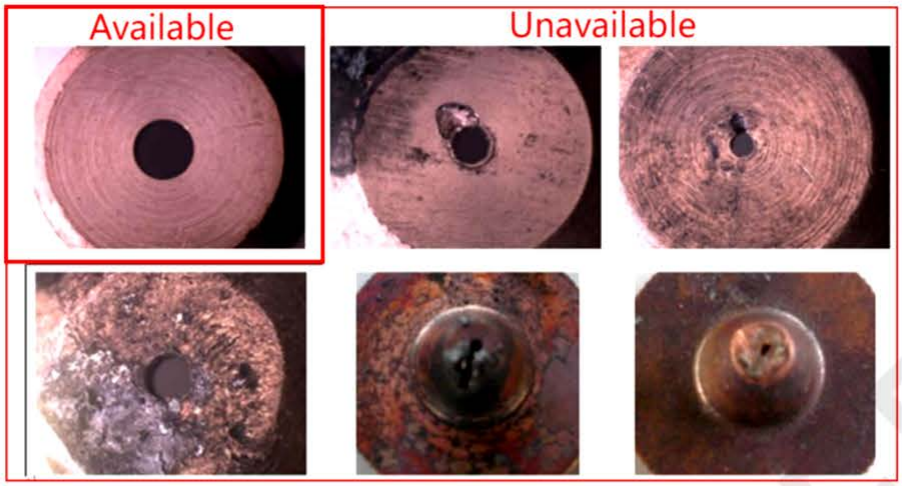
## 4.2 Laser cutting principle and debug method

Laser cutting uses a focused high-power-density laser beam to irradiate the workpiece, causing the irradiated material to rapidly melt, vaporize, ablate, or reach its ignition point. Simultaneously, the high-speed airflow coaxial with the beam blows away the molten material, thus cutting the workpiece.
Laser cutting is one of the thermal cutting methods.

### 4.2.1 Copper Nozzle

**Identify the quality of nozzles**

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Check whether the small hole in the center of the nozzle is worn out and out of round. If there is iron slag on the contact surface of the copper nozzle, it will cause the cutting head to vibrate during use, and slag will be attached to one side of the workpiece when cutting. The copper nozzle cannot be used if it changes color.

**Copper Nozzle Selection**

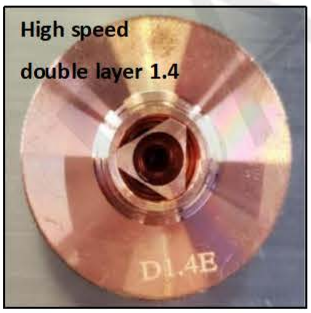
Selection of Nozzles for 6000W Cutting

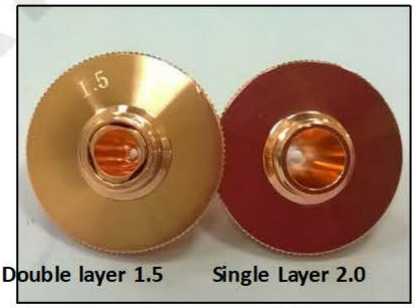
1. Oxygen Cutting for Carbon Steel

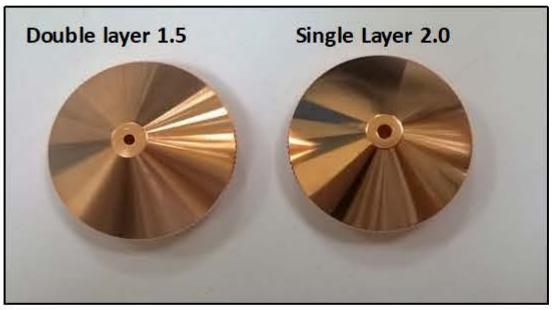
- Use double-layer nozzles.
- For thicknesses between 1-12mm, generally use a Φ double-layer 1.2 nozzle.
- For thicknesses between 14-18mm, generally use a double-layer Φ st1.2 or st1.4 nozzle.
- For thicknesses between 20-22mm, generally use a double-layer Φ st1.4 or st1.6 nozzle.

2. Nitrogen Cutting for Stainless Steel

- Use single-layer nozzles.
- For thicknesses below 6mm, use a single-layer 3.0 nozzle.
- For thicknesses between 8-12mm, use a single-layer 5.0 nozzle.
- For thicknesses between 14-20mm, use a single-layer 5.0-8.0 nozzle.

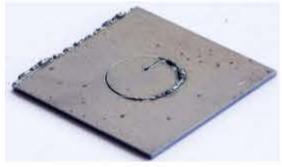


## 4.2.2 Oxygen Cutting Process for Carbon Steel

### Carbon Steel Cutting

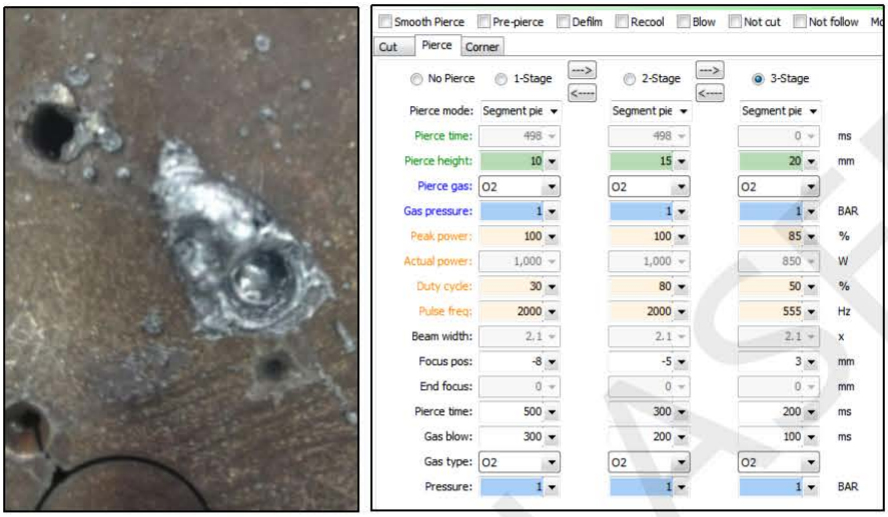
| Cut surface | Possible Reasons | Adjustment Methods |
|---|---|---|
|  | 1.Incorrect nozzle selection: Nozzle size too large.<br>2.Focal point mismatch: Consider further increasing positive defocus.<br>3.Cutting speed too slow. | 1.Replace the nozzle: Use a nozzle with a smaller diameter.<br>2.Increase the positive focus: Continue increasing positive defocus, and consider raising the nozzle height. Correspondingly, increase the gas pressure.<br>3.Increase the speed appropriately: Adjust speed to a reasonable level and observe the improvement in slag adhesion. |
|  | 1.Incorrect nozzle selection: Nozzle size too large.<br>2.Focal point mismatch: Consider further increasing positive defocus.<br>3.Cutting speed too slow, gas pressure too high. | 1.Replace the nozzle: Use a smaller diameter nozzle.<br>2.Increase the positive focus: Continue increasing positive focus.<br>3.Increase speed appropriately: Adjust speed to a reasonable value and observe improvements in slag adhesion. |
|  | 1.Cutting speed too fast.<br>2.Focal point mismatch: Consider further increasing positive defocus.<br>3.Damaged copper nozzle or misaligned optical path. | 1.Reduce the cutting speed appropriately.<br>2.Increase positive focus.<br>3.Replace the copper nozzle and adjust the optical path. |
|  | 1. Cutting speed too slow.<br>2. Protective lens is dirty.<br>3. The gas is impure and the pressure is too high. | 1. Increase speed appropriately.<br>2. Clean or replace the protective lens.<br>3. Replace the gas cylinder and reduce cutting pressure. |
|  | 1. Cutting speed is too fast.<br>2. The air pressure is too small.<br>3. Wrong focus. | 1. Reduce cutting speed.<br>2. Increase gas pressure appropriately.<br>3. Adjust the focus to the positive defocus. |

### Carbon Steel Piercing

1. The first-level piercing height must be at least 10mm, and the piercing pressure should not be lower than 0.6 BAR to avoid back splatter on the protective lens.
2. Pay attention during piercing, stop immediately if there is an explosion or failure to penetrate. Adjust the

piercing parameters and restart; continuing without stopping may cause back splatter that can damage the protective lens.

3. When using single-level piercing, avoid setting too high piercing power.
4. Adjust the piercing pressure when changing nozzle diameters to prevent explosions.
5. For long piercing times, reduce the power accordingly; if power is increased, reduce the piercing time.

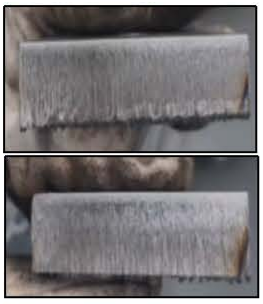



As shown in the figure, it is necessary to observe at which stage the blast hole appears during piercing.
There were no holes in the piercing at the beginning, but there were holes in the second stage. It is necessary to examine several important factors that affect perforation in the second stage.
Check whether the air pressure, power, duty cycle, and focus of the piercing are too large, and whether the residence time is too long, causing holes to explode. The duty cycle can be appropriately reduced or only the dwell time can be reduced.

## 4.2.3 Nitrogen Cutting Process for Stainless Steel

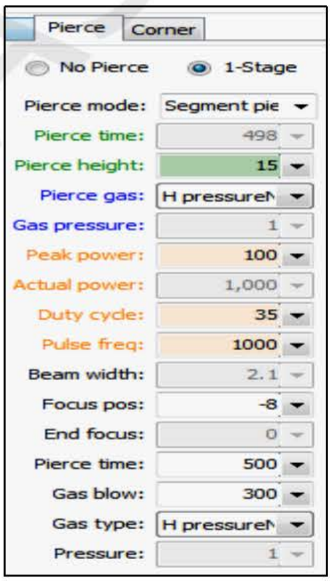
**Stainless Steel Cutting**

| Cut surface | Possible Reasons | Adjustment Methods |
|---|---|---|
|  | 1. Negative focal point mismatch: Excessive negative focus causes plasma formation, blue light, and layer separation.<br>2. Wrong nozzle selection: the nozzle diameter is too small.<br>3. Air pressure feed error: Air pressure feed is too large. | 1. Reduce the negative focal value: Decrease negative focus, keep other parameters the same, and observe the results.<br>2. Replace with a larger diameter nozzle.<br>3. Reduce gas pressure feed appropriately. |

| | | |
|---|---|---|
|  | 1. Incorrect nozzle selection: Nozzle size too small.<br>2. Negative focus mismatch: Consider further increasing negative focus.<br>3. Insufficient gas pressure: Not enough pressure for proper cutting. | 1. Replace the nozzle: Use a larger nozzle to increase gas flow.<br>2. Increase negative focus: Continue increasing negative focus to reach the bottom of the cutting section.<br>3. Increase gas pressure further. |
|  | 1. Nozzle is too small, insufficient airflow.<br>2. Focal point too high, cannot remove burrs.<br>3. Speed too slow. | 1. Replace with a suitable nozzle.<br>2. Adjust focus towards the negative direction.<br>3. Increase speed appropriately. |

**Stainless Steel Piercing**

1. High-power piercing differs from low-power, incorrect piercing techniques can often damage the protective lens.
2. Piercing height must be at least 10mm, and pressure not lower than 2 BAR, otherwise back splatter will easily reach the protective lens.
3. Piercing thick stainless steel requires penetration after 2 stages for best stability.
4. Piercing must penetrate instantly, using high-frequency breakdown to reduce slag on the surface.
5. If there is excessive slag on the stainless steel surface, use the slag removal function to reduce its thickness.




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# Chapter 5 Operation

## 5.1 Control Panel Overview

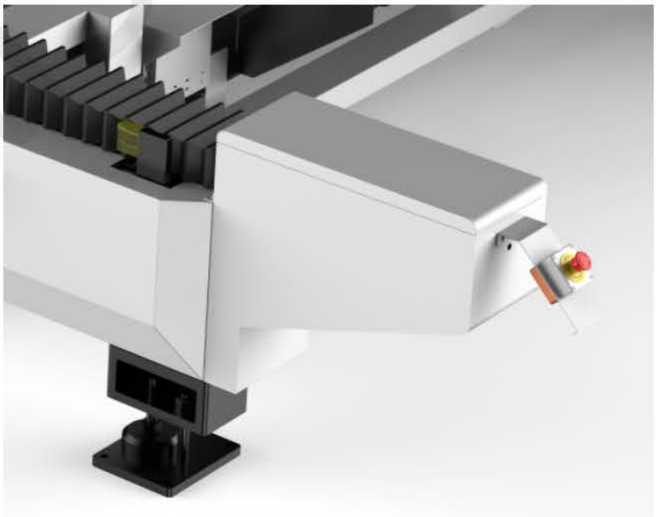
The operation units include the main machine control panel and the handheld controller. The functions of the main control panel and handheld controller are explained below.

### 5.1.1 Main Machine Control Panel

The frame of the main operation panel of the equipment is shown in the figure below. It is mainly used to complete some control operations related to power on/off, emergency stop, and moving magnification of the machine. From left to right they are: emergency stop button, main power switch, and laser power supply.



The end of the equipment is equipped with an emergency stop switch.



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## 5.1.2 Handheld Controller

The manual pulse generator is flexible and can be extended to the operating terminal. Connect it to the BOCHU controller through wireless signals.

The two common functions of the manual pulse generator are as follows:separation cutting and changing the origin position of the cutting program.

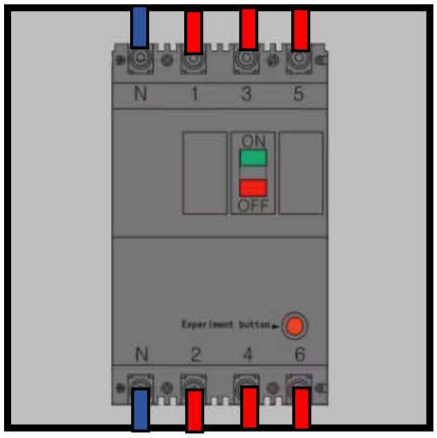
| button | Button content | button | Button content | picture |
|---|---|---|---|---|
| Start | Start and Continue Cutting | Pause | Pause |  |
| Continue | Continue cutting | Stop | Stop cutting | |
| Blow | Release/Stop gas button | Follow | Cutting head up and down follow button | |
| Shutter | Laser enable/disable button | Aiming | Red light button (IPG) | |
| PT LOC | Return to the position after clicking the stop button | Laser | Press once to emit the laser. Hold down to continuously fire the laser | |
| Back ← | The cutting head moves back (in paused state) | Forward → | Cutting head moves forward (in paused state) | |
| Edge Seek | Find the edge of the cutting workpiece | Fn | Function keys | |
| Dry Cut | Walk once through the selected graphics | Zero | Back to the starting point | |
| High | Press and hold the direction keys (X, Y, Z axis) to move quickly | Step | Press and hold the direction keys (X, Y, Z axis) to move according to the set moving distance | |
| WC | Clockwise rotation | WC | Counterclockwise rotation | |
| → | Cutting head moves to the right | ← | Cutting head moves to the left | |
| ↑ | Crossbeam moves backward | ↓ | Crossbeam moves forward | |
| Z↓ | Cutting head moves upward | Z↑ | Cutting head moves downward | |
| K1-K6 | Customizable buttons | Fn+K1-k6 | Customizable buttons,Hold down the Fn key and then press the K key | |

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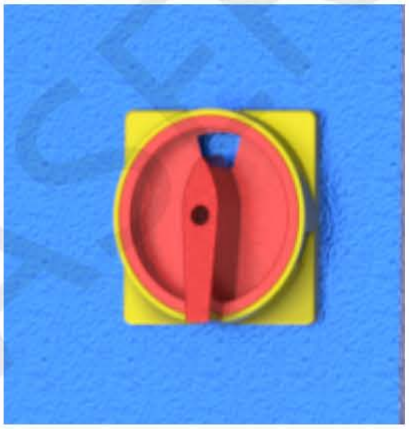
| | | | | |
|---|---|---|---|---|
| | | | for the action to take effect | |
| The K key is a customizable button, and its function may vary. The function of the K key is for reference only; please refer to the actual equipment for accurate functionality. | | | | |

## 5.2 Machine startup sequence

(1) Turn on the main power switch: Activate the main electrical cabinet, voltage stabilizer, and the machine's main power switch.



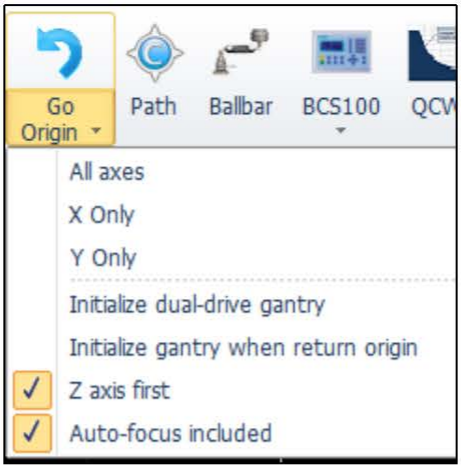


(2) Release the emergency stop knob: Twist the emergency stop knob to release it, then press the computer power button (machine control system) and wait for the control system to start up.



(3) Enable servo motors: Click the servo button, launch the cutting software, and inspect the machine for any foreign objects before executing the "Return to Origin" operation.




(4) Start the laser and chiller power: Turn on the laser and chiller power. Allow the chiller to run for 15 minutes until the water temperature exceeds 20°C. For lasers of 6000W or higher, ensure an air-conditioned room is installed and activate the dehumidification function. Check the cutting head water pipes for leaks and confirm the chiller's water level and operating status.




(5) Check gas pressure: Verify the pressure of the gas valves based on the required cutting gas. Open the gas regulator valves and check the pressure of the cutting auxiliary gases. Use the handheld controller to perform a gas purge to ensure proper gas flow.

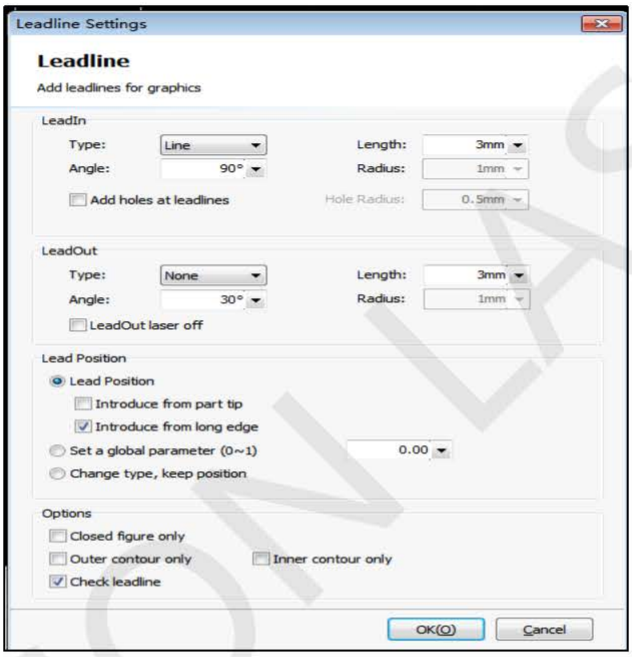




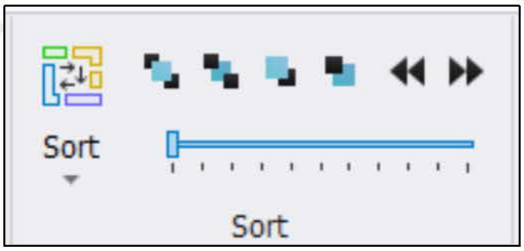
## 5.3 Cutting Procedure

Step 1: Open the cutting software "Cypcut" and click on "File" to select the graphic that needs to be cut.

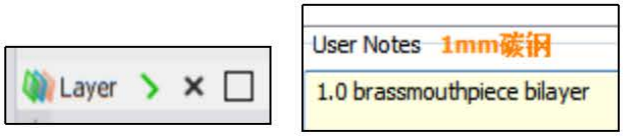
Step 2:Select the graphic and add a leader line by clicking "LeadLine" with a straight line type and a length of 5mm (modifiable as needed), without changing the angle. Choose "None" for the leader line position, which can be introduced from the long edge (the edge of the graphic), and click OK. After adding the leader line, if it extends outside the graphic, click on the icon below the leader line "Identify Inner/Outer Contour" to move the leader line inside the graphic. Alternatively, perform an "Inner" cut to transform the leader line position.




Step 3: Select the graphic and click “Sort.” A sorting method will pop up. Click “OK,” and the system will automatically arrange the sequence of the two graphics.




Step 4: Click on “layer”, retrieve the parameters corresponding to the material thickness, check the size of the corresponding copper nozzle, and click “OK.” Replace the copper nozzle on the cutting head.



Step 5: After replacing the copper nozzle, perform capacitance calibration and air pressure calibration.

Step 6: Click on "CNC" to illuminate "BLF", select the appropriate cutting gas, and then click on "Air Blowing" confirm that gas is being expelled.

Step 7: Select the graphic, move the cutting head to the cutting position, and click “Frame” to observe the area occupied by the graphic.

Step 8: Once confirmed, click the “Start” button to begin processing

Note: First-time users should keep the handheld controller nearby with their hand over the “Pause” button. If any issues arise, press the “Pause/Stop” button immediately.

## 5.4 Shutdown steps

- After cutting is completed and no secondary cutting is required, press the emergency stop button on the console→turn off the laser key switch→turn off the chiller rotary switch
- Close all gas bottles and valves, use software to blow out the pipeline gas, and turn off the air compressor and cold dryer
- Turn off the servo button on the console；save the cutting parametersand the cutting graphics processing task
- Close the cutting software → close windows normally→turn off the computer button；
- Turn off the main power "circuit breaker" of the electrical cabinet

Note：

- Please strictly follow the operating steps described above to ensure the normal operation of the machine and equipment and avoid malfunctions and dangers.
- If the on-site environment is harsh, such as extremely low temperature and other special circumstances, be sure to consult the sales engineer.

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# Chapter 6 Maintenance

## 6.1 Overview

In order to ensure the normal use of the laser cutting machine , the equipment must be maintained on a daily basis. Since the entire machine tool is composed of high-precision components, special care must be taken during daily maintenance, and the operation procedures of each part must be strictly followed. Maintenance must be performed by a dedicated person, and rough operation is prohibited to avoid damage to components.

**Spare parts that users should always have on hand**

A. Isopropyl alcohol: purity 99.7%, water less than 0.3%, capacity 500ml per bottle.
B. Dust-free cotton swabs: 1 pack (100 swabs/pack).
C. Alcohol: 500ml, purity above 99.5%.
D. Lens cleaning paper: 1 piece.
E. Compressed air: 1 bottle.
F. Rust remover: 1 bottles.
G. Spray butter: 1 bottles.
H. All-purpose water: 1 bottle.
I. Multimeter: one.

### 6.1.1 Maintenance of the gas circuit

Check the gas pipeline for leaks every day; check whether the drain valves of the air compressor, gas tank, dryer, and filter are draining normally, and check whether all drainage is normal.

### 6.1.2 Maintenance of waterway

| Machine Name: | Machine Model: | Machine Number: |
|---|---|---|
| maintainance Time | Maintenance content | Maintenance target |
| daily | 1. Check whether the temperature setting of the chiller is normal (set temperature 20℃-22℃) | Ensure that the cooling water temperature supplied to the laser is normal |
| | 2. Check whether the water circuit seal, water temperature and water pressure of the chiller meet the requirements. | Ensure normal operation of equipment and prevent water leakage |
| | 3. Keep the working environment of the chiller dry, clean and ventilated | Conducive to the good operation of the chiller |
| | 4. Observe the liquid level indicator tube at the front of the chiller to | Prevent the water pump from burning |

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| | | |
|---|---|---|
| | make sure the water level in the water tank is above 90% of the water tank height. | out due to idling |
| | 5. Make sure the fixing screws of the water tank cover are tightened and the water filling cap of the water tank is tightened. | Prevent dust and debris from entering the water tank |
| Every month | 1. Use detergent or high-quality soap to remove dirt on the surface of the chiller. Do not use benzene, acid, abrasive powder, steel brush, hot water, etc. to clean. | Ensure the surface of the chiller is clean |
| | 2. Open the door panel of the chiller chassis, use a high-pressure air gun to blow the cooling fins of the condenser radiator to blow off the dust attached to the cooling fins. | Prevent poor cooling and burning of water pumps and compressors |
| | 3. Use a high-pressure air gun or rag to clean the water pump heat sink and other parts, and use a high-pressure air gun to blow away the dust on the cooling fan behind the water pump and the debris in the chassis. | Prevent poor heat dissipation and increase the service life of each component |
| | 4. Check the water quality of the water tank and follow up | Good water quality can ensure the normal operation of the laser |
| | 5. Check whether there is water leakage in the chiller pipeline | Ensure that there is no water leakage in the chiller |
| Every three Month | 1. Check electrical components (such as wiring terminals, switches, etc.) and remove dust | Ensure the surface cleanliness of the electrical parts of the chiller to extend its service life |
| | 2. Replace circulating water (distilled water) | Prevent impurities in the water from clogging the circulating water system, which can seriously affect the cutting effect and even burn out the optical components. |
| | 3. Clean the filter; (Note: Use high-pressure water to clean the filter, please be careful during the cleaning process) | Prevents the filter from becoming clogged and not functioning properly |
| | 4. Clean the chiller water tank | Prevent impurities in the water tank from entering the water circulation system |

**Note:**

a. Before adding water, make sure the chiller is turned off and the drain valve at the bottom of the chiller is closed. If there are visible impurities or foreign objects, please do not rush to add water. Clean the water tank first before adding water.

b. When adding water, please be careful to avoid water leakage outside the chiller, causing rust on the surface of the chiller or dirt in the environment;

c. Use high-pressure water to clean the filter. Please be careful during the cleaning process;

d. It is strictly forbidden to place the chiller in an environment below 5℃ and above 40℃;

e. The chiller must be kept ventilated smoothly during use: ensure that there is no obstruction within 1000mm on the upper air blowing side and the air inlet side is

No obstruction within 1000mm;

f. The connecting water pipe between the chiller and the cutting machine is generally 10 meters in one-way length

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and the longest length shall not exceed 25 meters.

## 6.1.3 Maintenance of exhaust dust removal system

The exhaust dust removal system is an auxiliary component of the laser tube cutting machine. Its main function is to timely and effectively remove the dust generated by laser cutting, protect the important precision parts of the machine tool from dust pollution, and protect the workshop environment. The maintenance of the exhaust dust removal system can be carried out according to the maintenance instructions of the exhaust fan, and the details of each maintenance should be recorded in the maintenance registration form:

| Machine name: exhaust fan | Machine Model: | Machine Number: |
|---|---|---|
| Maintenance period | Maintenance content | Maintenance target |
| daily | Remove stains and dirt from the exhaust fan surface | Keep the exhaust fan clean |
| Every 7 days | Clean the inside of the exhaust fan | Ensure smooth ventilation and good effect |

Note: The exhaust dust removal system is optional. Equipment equipped with the exhaust dust removal system requires this maintenance.

## 6.1.4 Maintenance of voltage stabilizer

A voltage stabilizer is a power supply circuit or power supply device that automatically adjusts the output voltage; its main function is to stabilize the power supply voltage that fluctuates greatly or does not meet the requirements of electrical equipment within its set value range, so that various circuits or electrical equipment can work normally under the rated working voltage. In the laser tube cutting machine, the voltage stabilizer mainly provides a stable working voltage that meets the requirements for the laser to avoid damage to the laser or other electrical equipment due to unstable voltage or voltage that does not meet the requirements. The use of voltage stabilizers in laser tube cutting machines is particularly important, and the maintenance of voltage stabilizers is also essential; please refer to the table below for specific maintenance and operation methods and strictly follow them, and record the situation of each maintenance.

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| Machine name: voltage stabilizer | Machine Model: | Machine Number: |
|---|---|---|
| Maintenance period | Maintenance content | Maintenance target |
| daily | 1. Check whether the load current is within the allowable range | Ensure the normal operation of the machine |
| | 2. Remove stains and dirt from the machine surface | Ensure the machine surface is clean |
| | 3. Check the three-phase voltage stability | Ensure normal power supply to machine tools and lasers |
| per month | 1. Thoroughly clean all parts of the voltage stabilizer, especially the brushes, exposed parts of the contact voltage regulator, brush sliding guide rails and speed transmission parts. They must be cleaned and wiped dry with gasoline and cotton cloth; | Ensure good heat dissipation of accessories and extend the service life of accessories |
| | 2. Replace worn or damaged brushes; | Ensure the stability of three-phase voltage |
| | 3. Adjust the tightness of the chain or the clearance of the screw, leaving a little room for movement. | Ensure the normal operation of the machine |

**★★★Notes:**

a. The input voltage is within the allowable range. If the output voltage is unstable, check it immediately;

b. The voltage stabilizer should be installed in a clean, dry, ventilated and cool place to avoid overheating during use. Remember: Although the machine can be installed in an environment with a room temperature of 40°C, long-term use at such high temperatures will cause changes in semiconductor parts and cause permanent damage, thereby reducing its reliability. For safety reasons, it is recommended to use it at a room temperature of 25°C.

c.Maintenance must be carried out with the power off.

Note: The voltage stabilizer is configured according to the power size, and the equipment equipped with the voltage stabilizer needs to do this maintenance.

## 6.2 Laser maintenance

The specific maintenance of the laser is mainly based on the manual of the selected laser.

**A) Antifreeze**

When the use environment is often out of power and the conditions for daily coolant draining are not met, antifreeze must be used. The base liquid of antifreeze is generally composed of alcohol and water, and requires high boiling point and flash point, large specific heat and conductivity, low low-temperature viscosity, non-foaming, and no corrosion to metal parts, rubber hoses, etc. When selecting or mixing antifreeze, its freezing point should be 5°C lower than the lowest temperature of the use environment. Use a professional brand of special antifreeze:

It is recommended to choose the CLARIANT brand. There are two types of antifreeze suitable for laser systems:

1) Antifrogen® N glycol-water type (industrial product, toxic to human body)

2) Antifrogen®L Propylene Glycol-Water Type (food grade, harmless to human body)

Note: Any antifreeze cannot completely replace deionized water, cannot be used for a long time throughout the year, and the volume used should account for 10%-20% of the water tank volume, and cannot exceed 20%. After winter, the pipeline must be cleaned with deionized water or purified water, and deionized water or purified water must be used as a coolant again.

**B) Anti-condensation**

- Since lasers have high requirements for the use environment, in order to protect the laser in summer and prevent condensation caused by the large temperature difference between the laser and the external environment, the following maintenance is required when using the laser in summer:
- Provide a separate air-conditioned room for the laser to improve the external working environment of the laser and keep the laser working in a dry and constant temperature environment;
- It is recommended that customers purchase a temperature and humidity meter and place it in the air-conditioned room of the fiber laser to monitor the temperature and humidity in the air-conditioned room;
- Adjust the temperature of the chiller according to the ambient temperature, set the high temperature to 28-32℃ and the low temperature to 25-29℃;

**C) Strictly follow the power on and off sequence of the fiber laser machine.**

- **Boot sequence**

(1) Turn on the laser main power switch and let the chassis air conditioner or dehumidifier run for more than 30 minutes;

(2) After turning on the laser for 30 minutes, turn on the chiller power switch.

- **Shutdown sequence**

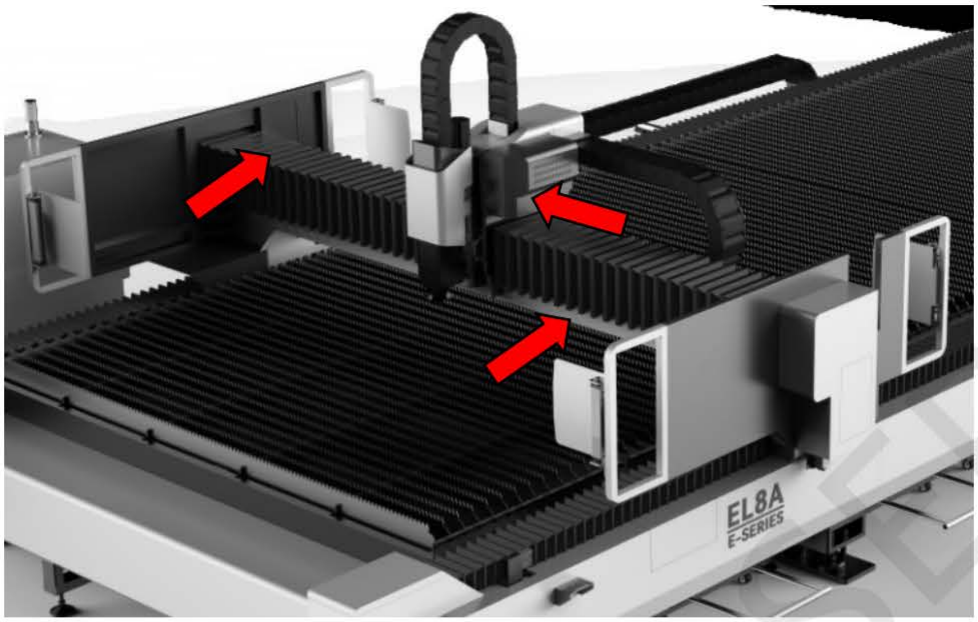
(1) Turn off the power switch of the chiller;

(2) Turn off the laser power switch.

## 6.3 Cutting Machine Maintenance

Cutting metal sheets generates a large amount of dust. To ensure the machine's long-term stable operation, internal maintenance is also essential. Clean the dustproof cover, sheet metal, and beam parts monthly.(Power-off operation)

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**When the Machine Is Not in Use for a Long Time**

When the laser cutting machine is not in use for an extended period, apply grease to the moving parts of the machine, wrap them in anti-rust paper, and regularly inspect other parts for rust. Remove rust and apply anti-rust treatment to rusty parts (if possible, cover the machine with a dustproof cover), and regularly clean and inspect the machine.

## 6.4 Cleanliness of the Electrical Control Cabinet

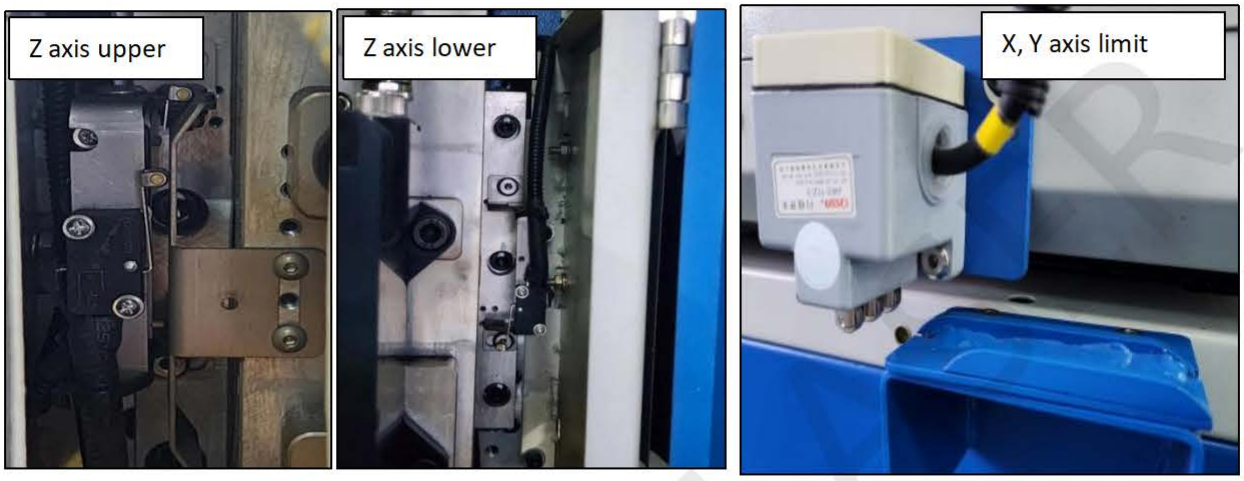
Over time, dust can accumulate inside the electrical cabinet. Use a clean cloth to wipe down the cabinet components monthly, and every three months, check if the wiring is loose. Each plug-in connector should be checked thoroughly. (This must be done with the power off).





## 6.5 Limited cleaning

The sensor limits of the X-axis, Y-axis and exchange platform every week to see if they are normal . Touch the sensor with an iron block to see if the light is on and if the computer monitor alarms. Clean the sensor with cotton cloth to avoid dust covering it for a long time, which may cause insensitivity.




## 6.6 Cleaning of guide rails and gear racks

As one of the core components of the equipment, the guide rail and gear play a guiding and supporting role. In order to ensure that the machine has a high processing accuracy, the guide rail is required to have a high guiding accuracy and good movement stability. In order to make the machine work normally and stably and ensure the processing quality of the product, the daily maintenance of the guide rail and gear should be done carefully.

Notice :

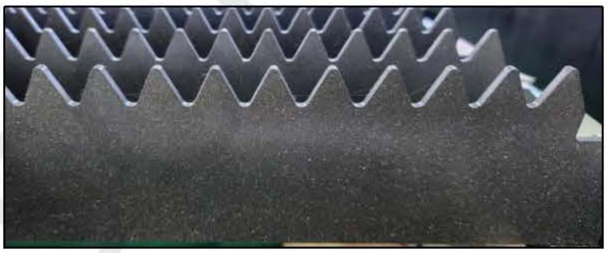
Please prepare dry cotton cloth and lubricating oil to clean the guide rail. Cleaning of linear guide rail: First move the laser head to the far right (or left), find the linear guide rail, wipe it with dry cotton cloth until it is bright and dust-free, add a little lubricating oil, and slowly push the laser head left and right several times to evenly distribute the lubricating oil. Cleaning of roller guide rail: Move the crossbeam to the inside, open the end covers on both sides of the machine, find the guide rail, wipe the contact area between the guide rails and the rollers on both sides with dry cotton cloth, then move the crossbeam and clean the remaining area.

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## 6.7 Rack

Long-term use of racks for cutting will inevitably cause damage, which will cause unevenness when placing the plate. The height of the rack support will drop and the lower limit alarm will easily occur. Severely damaged racks will affect the cutting effect and cause slag on one side. Severely damaged racks need to be replaced.



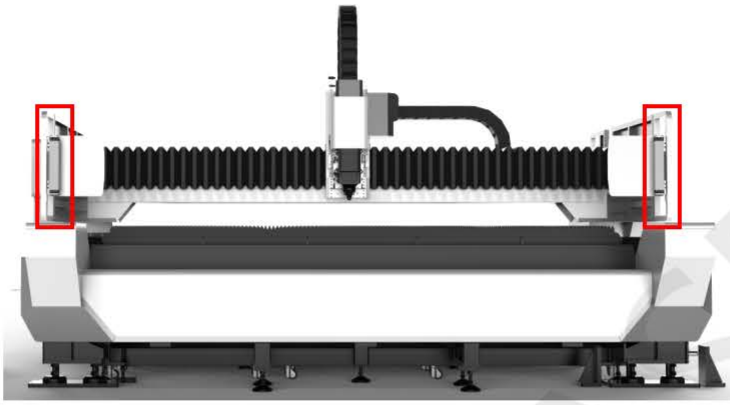
## 6.8 Inspection of machine tool foot cup

Every three months, check whether all the foot cups of the twisting machine tool are stable, twist the nuts on the foot cups to check whether they are loose, and tighten them with a spanner if they are loose.



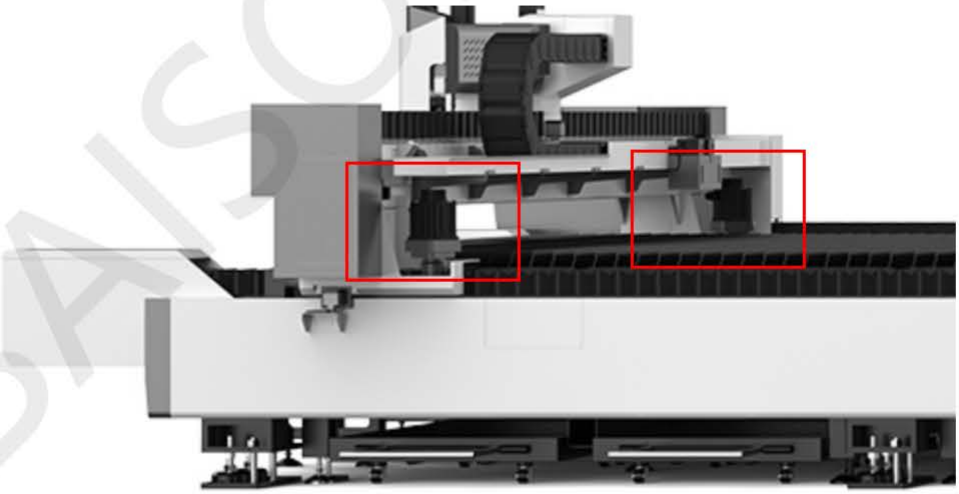
## 6.9 Inspection of gratings

Clean the grating sensing area according to the usage of the equipment. Prevent dust accumulation from becoming too thick and affecting the sensing.



## 6.10 Motor Cleaning

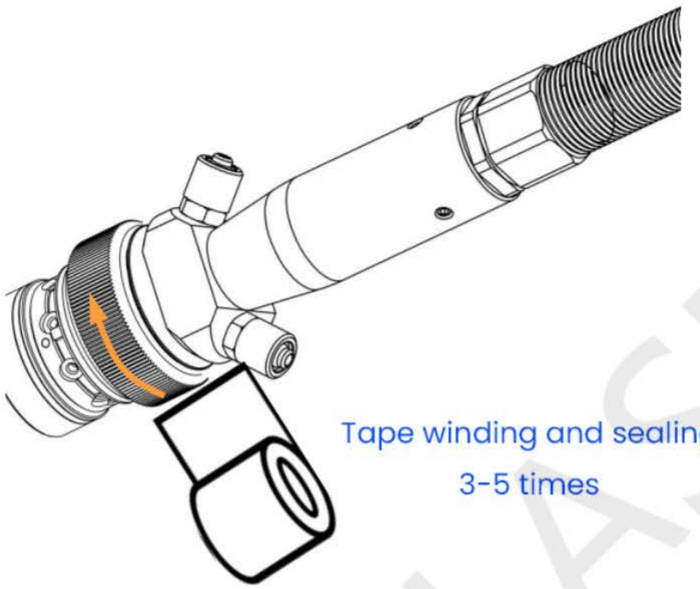
The motors on both sides of the X-axis will accumulate dust with the frequency of equipment use. After turning off the power supply, clean them with a clean cloth.



## 6.11 Cutting head

Clean the appearance of the cutting head every day. After a long period of cutting, clean it with a clean rag and then wrap the cutting head with masking paper to prevent smoke and yellowing. Wipe the QBH (fiber interface)

and wiring port of the cutting head with a clean cloth and then wrap it with masking paper. Pay attention to changing the masking paper on the lower protective lens door frequently to keep it clean and avoid dust from entering the lens when changing.

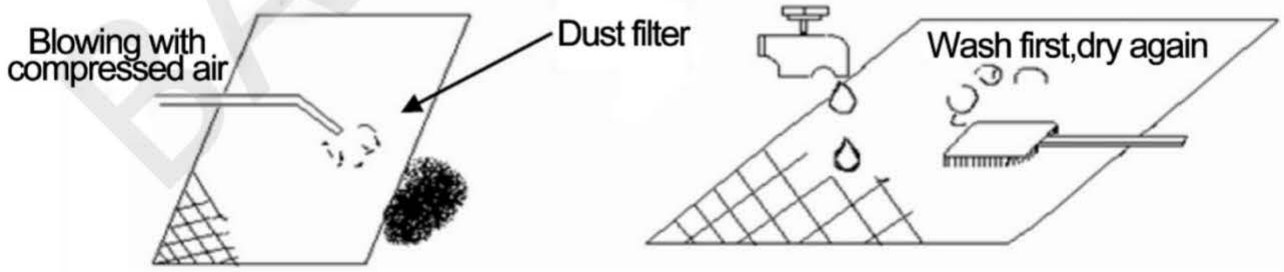



## 6.12 Chiller maintenance requirements

Note: Equipment maintenance must be stopped first , cut off the power supply , and wait for 3 minutes before proceeding , otherwise there is a risk of electric shock. When the ambient temperature is lower than 2℃ and the machine is shut down for a long time, the internal water must be drained.

### 6.12.1 Dust protection in summer

In summer , please clean the equipment condenser and dust filter in about 15 days , as shown in Figure 1.



## 6.12.2 Antifreeze in winter

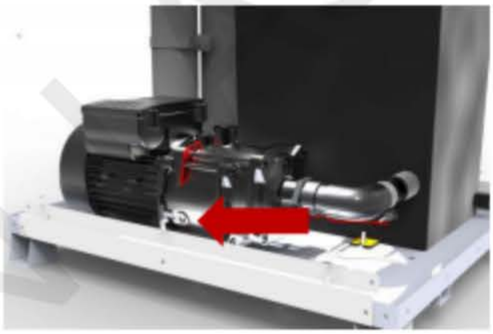
When the equipment is transported or not used for a long time , the water in the water tank should be drained through the drain valve , and at the same time, loosen the drain plug under the water pump to drain out the remaining water in the water pump , as shown in Figure 2 .

1. First open the drain valve and drain the water.



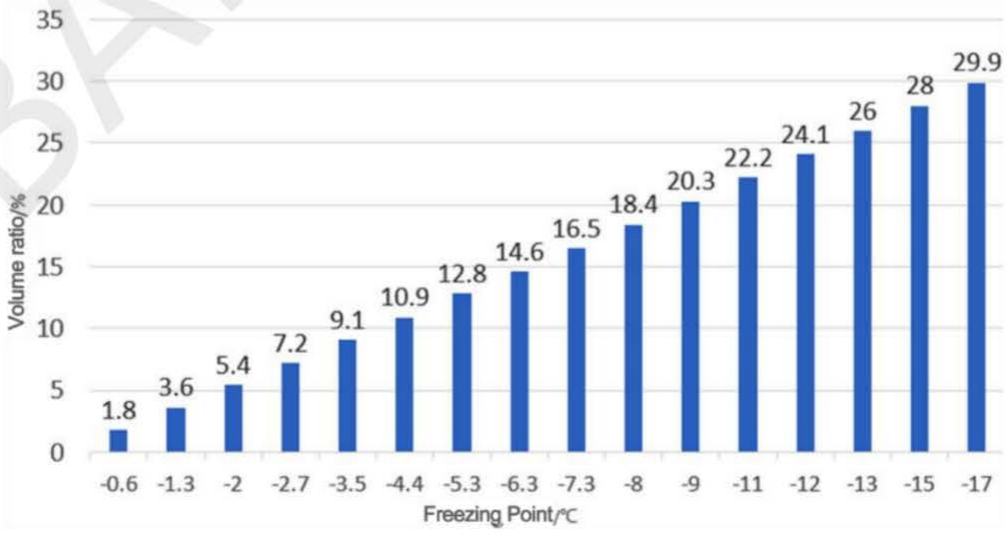


2. Unscrew the water pump exhaust screw



3. Remove the screws at the bottom of the water pump

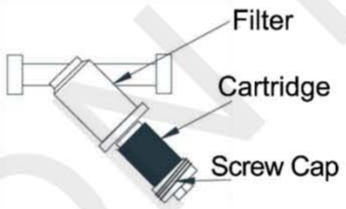
If the ambient temperature at night is lower than 2℃ , It is recommended that customers do not stop the machine or add antifreeze . The volume ratio of ethylene glycol is selected according to Table 1. The freezing point to be prevented is equal to the ambient temperature of the equipment minus about 5°C. When the daily average temperature is higher than 5℃ , replace the water containing antifreeze with softened water.



## 6.12.3 Daily maintenance

The working scene of the fiber laser chiller is very harsh . In order to ensure the good performance of the equipment and extend the service life of the equipment , the equipment needs to be maintained . The maintenance frequency is once a week . The maintenance work includes but is not limited to the following aspects:

- Check the condenser , check whether the air duct is blocked by foreign matter , and whether the air inlet and outlet around the equipment is unobstructed;
- Clean the condenser and clean the dust filter;
- Check whether there are foreign objects in the refrigerant and whether microorganisms breed, etc. Generally , the refrigerant should be replaced every 15 to 20 days . The refrigerant must be pure water, distilled water or high-purity water;
- Check whether the water pipe joints are loose and whether the water pump is leaking;
- Check the water tank and clean the dirt deposited inside the water tank;
- Regularly clean the filter in the waterway of the equipment . Generally , the filter should be cleaned once every 7 to 10 days . The commonly used filters of our company are as shown in Figure 3. The Y-type filter is placed at the water suction port of the water pump or the water outlet of the equipment ;




- Test insulation resistance , insulation resistance ≥5MΩ;
- Check the ground resistance , ground resistance ≤ 1 Ω;
- Conduct a capacitance test on the capacitors of the compressor and fan .
- If the capacitance decay exceeds 10%, the capacitors need to be replaced.

## 6.12.4 Hazard tips for inadequate daily maintenance

- If the dust filter or condenser is not cleaned in time, the heat exchange efficiency will be reduced, the heat will accumulate, causing the fluorinepressure to rise or even the high pressure alarm, resulting in the chiller can not operate normally. Running under this condition for a long time will seriously affect the service life of the compressor.
- If the filter is not cleaned in time, the dirty blockage of the filter will reduce the water flow, cause the water temperature to rise, and even cause the laser to alarm at high temperature, which will not work normally. If non-pure water is added or the refrigerant is not replaced in time, it may lead to the blockage of the water channel, reduce the water flow, cause the flow alarm of the chiller or the high temperature alarm of the laser, and make the chiller unable to work normally. In severe cases, it may cause the plate heat exchanger to be completely blocked, affecting the cooling capacity of the chiller, resulting in the chiller not working normally. If antifreeze is not added in time in winter, it will cause damage to water pumps, lasers, etc.

- If the capacitor capacitance of the compressor or fan is attenuated by more than 10%, the compressor or fan will be damaged.

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# Chapter 7 Troubleshooting

## 7.1 Cutting machine failure analysis

| Abnormality / failure | Reason | Workaround |
|---|---|---|
| The power is on, but the power indicator is off | No external input power | Connect power to the main power source or circuit breaker distribution box |
| | There is a problem with the main power contactor | Replace the contactor or check the circuit |
| | The main power switch, key switch, and emergency stop button have poor contact or are damaged. | If necessary, replace the damaged switch, or the damaged switch wires. |
| No auxiliary gas output during parts processing | Insufficient Gas pressure | Check the Gas pressure |
| | The solenoid valve is damaged or the circuit is damaged | Check the solenoid valve electrical circuit and solenoid valve |
| There is an abnormal sound when the axis is running | No lubrication of the shaft moving parts | Check the fluid level of the central lubrication pump and add lubricating oil if necessary. |
| | There is interference between moving parts and fixed parts | Check the safety of the running path of moving parts |
| The cutting pattern does not match the drawing size | Program input error | Check the program by drawing |
| | Positioning accuracy impact | Check machine tool accuracy, software and servo motor pulse equivalent |

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## 7.2 Laser Power Supply Failure Analysis

| Abnormality/failure | Reason | Workaround |
|---|---|---|
| The laser power supply does not work and the panel indicator light is off | The power supply is abnormal and the plug is loose. | Replace the plug |
| | Power not connected | Check the power supply cables |
| The power supply is normal, but the main circuit is not working. | Is the water circulation normal ? | Turn on circulating water |
| | Emergency stop, key normal? | Return to normal position |
| Cutting gap is too wide | The focus is not adjusted properly | Refocus |

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## 7.3 Cooling system failure analysis

| Abnormality/failure | Reason | Workaround |
|---|---|---|
| No display after power on | The power supply phase is broken or the voltage is low | Power supply on demand |
| | Thermostat failure | Check according to the thermostat setting method and replace the thermostat if necessary |
| The compressor cannot start or suddenly stops during operation ( the overload light is on ) | Improper low pressure control setting | Reset low pressure value |
| | High-voltage controller actuated ( maybe the ambient temperature is too high) | Find out the cause and fix it , then reset the button |
| | Refrigeration pipe is blocked or leaking | Ask professionals to handle |
| | The pressure controller is set correctly and the protection is activated | Low pressure protection : lack of refrigerant , check for leaks , eliminate them and add refrigerant<br>High voltage protection : The condenser is dirty or the condensing fan is broken. After repairing , press the red button of the voltage controller to reset it. |
| | Thermostat failure | Repair or replace the thermostat |
| Compressor frost | The temperature setting is too low or the thermostat is broken | Adjust the temperature or replace the thermostat |
| | The expansion valve is opened too wide, there is too much refrigerant in the system , the expansion valve temperature sensor is improperly positioned or not fastened | Adjust the valve opening , remove excess refrigerant , and rewrap the temperature sensing package as required . |
| | The evaporator is dirty and the heat exchange efficiency is low | Cleaning the evaporator |
| During operation, the flow indicator lights up and the buzzer sounds an alarm | Insufficient water flow, flow switch actuated | Is the water pump empty? |
| | Water filter is dirty and clogged | Cleaning the water filter |

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| | | |
|---|---|---|
| Insufficient cooling capacity | The expansion valve is opened too large or too small | Adjust the expansion valve |
| | Insufficient or excessive refrigerant | Adjust to a reasonable value |
| | Too much air in the system | Re-evacuate and add refrigerant |
| | Refrigeration pipeline is dirty and blocked | Find out the cause and eliminate the fault |
| | Scaling on the water side of the evaporator (including plate type) | Cleaning the evaporator |
| | Condenser is too dusty | Cleaning the condenser |
| | Condensing fan failure | Repair or replace the fan |
| | Refrigerant leak | Check the leak point, repair it and add refrigerant |
| During operation, the water level indicator lights up and the buzzer sounds an alarm | Insufficient water level in the water tank | Drink enough water |
| | The water level probe is not in the correct position or the float is stuck | Adjust the probe and float |
| | Thermostat troubleshooting | Or replace the thermostat |
| Exhaust temperature is too high | The condenser is dirty and the fan air volume is insufficient. | Clean the condenser , repair or replace the fan |
| | The refrigeration system contains too much air | Re-evacuate and add refrigerant |
| | Low temperature pipe insulation layer damage | Repair insulation |
| | The heat load is too large and exceeds the design | Configure and use according to design requirements |
| | Ambient temperature is too high | Improve the use environment |
| Insufficient water pump flow | The water pump flow passage is partially blocked | Cleaning water pumps , water filters and pipes |
| | Water pump impeller damaged | Replace the impeller |
| | The flow resistance is too large and the total head is higher than the rated value of the pump | Check the cause , adjust the pipeline or replace the pump |

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